Potential Theory in the Complex Plane

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General Information

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Chapter I.

A Physical Motivation of Potential Theory

Potential theory has its origins in mathematical physics of the 19th century, namely in the study of gravity and the electromagnetic force. Let us take a look at electrostatics.

Consider a (negatively) charged body $K \subseteq \mathbb{R}^3$. The body is surrounded by an *eletric field* $\vec{E}$, i.e. the force acting on a test particle with the charge q at the position $\vec{x} = (x^1, x^2, x^3) \in \mathbb{R}^3 - K$ is $q\vec{E}(\vec{x})$. By Coulomb's law, a particle with the charge q_0 at the point $\vec{x}_0 = (x_0^1, x_0^2, x_0^3)$ induces the electric field

$$\vec{E}(\vec{x}) = \frac{1}{4\pi\varepsilon_0} \frac{q_0}{\|\vec{x} - \vec{x}_0\|^3} (\vec{x} - \vec{x}_0),$$

where ε_0 ist the *vacuum permittivity* and $\|\vec{x} - \vec{x}_0\|$ is the euclidean distance between $\vec{x}$ and $\vec{x}_0$. At the end of the 18th century, it was observed by Lagrange that there exists a scalar-valued function Φ, called the *potential of* $\vec{E}$, such that $\vec{E} = -\operatorname{grad} \Phi$, where $\operatorname{grad} \Phi(\vec{x}) = (\partial_1 \Phi(\vec{x}), \partial_2 \Phi(\vec{x}), \partial_3 \Phi(\vec{x}))^t$ is the gradient of Φ in $\vec{x}$ with respect to the standard basis of $\mathbb{R}^3$ and the euclidean inner product. Indeed,

$$\Phi(\vec{x}) = \frac{1}{4\pi\varepsilon_0} \frac{q_0}{\|\vec{x} - \vec{x}_0\|}.$$

This law has the advantage that the *work* W done by the electric field $\vec{E}$ when it moves a particle with charge q from a point $\vec{x}_1$ to a point $\vec{x}_2$ along the path $\gamma \colon [t_1, t_2] \to \mathbb{R}^3$ with $\gamma(t_1) = \vec{x}_1$ and $\gamma(t_2) = \vec{x}_2$, which by experiment is found to be

$$W = \int_\gamma q\vec{E}(\vec{x}) = q \int_{t_1}^{t_2} \langle \vec{E}(\gamma(t)), \dot{\gamma}(t) \rangle \, dt,$$

can be computed easily. Namely it holds

$$\langle \vec{E}(\gamma(t)), \dot{\gamma}(t) \rangle = -\langle \operatorname{grad} \Phi(\gamma(t)), \dot{\gamma}(t) \rangle = -(\Phi \circ \gamma)'(t)$$

so that $W = -q(\Phi(\vec{x}_2) - \Phi(\vec{x}_1))$. Now suppose that we have charges which are "continuously" distributed over the body K, i.e. there is a function $\rho\colon K \to \mathbb{R}$, the so-called *charge density*, such that

$$Q_B := \int_B \rho(\vec{x}_0)\, dx_0^1 dx_0^2 dx_0^3$$

yields the charge of the portion $B \subseteq K$. Then, the electric field surrounding the body K in the point $\vec{x} \in \Omega := \mathbb{R}^3 - K$ is given by

$$\vec{E}(\vec{x}) = \frac{1}{4\pi\varepsilon_0} \int_K \frac{\rho(\vec{x})}{\|\vec{x} - \vec{x}_0\|^3} (\vec{x} - \vec{x}_0)\, dx_0^1 dx_0^2 dx_0^3,$$

which has the potential

$$\Phi(\vec{x}) = \frac{1}{4\pi\varepsilon_0} \int_K \frac{\rho(\vec{x}_0)}{\|\vec{x} - \vec{x}_0\|}\, dx_0^1 dx_0^2 dx_0^3\,.$$

One can check that $\Phi\colon \Omega \to \mathbb{R}$ satisfies $\Delta\Phi \equiv 0$, where $\Delta := \sum_{i=1}^3 \partial_i^2$ is the *Laplace operator*, i.e. Φ is a harmonic function. Potential theory explains, in particular, how ρ can be recovered from (a suitable extension) of Φ; this leads to the differential form of *Gauss's law*.

Chapter II.

Harmonic functions

Let $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ be an open subset. Note that we suppose that $\mathbb{R}^N$ is endowed with the euclidean norm, i.e. for $x = (x^1, \dots, x^N)$ and $y = (y^1, \dots, y^N) \in \mathbb{R}^N$ we have

$$\|x\| := \Big(\sum_{i=1}^{N} |x_i|^2 \Big)^{1/2}, \qquad \langle x, y \rangle = \sum_{i=1}^{N} x^i y^i.$$

We denote

$$C(\Omega) := \{ f \colon \Omega \to \mathbb{R} \mid f \text{ is continuous} \},$$
$$C^k(\Omega) := \{ f \colon \Omega \to \mathbb{R} \mid f \text{ is } k \text{ times continuously differentiable} \}$$

and by $C^\infty(\Omega) := \bigcap_{k \geq 0} C^k(\Omega)$ we denote the space of smooth, i.e. arbitrarily often continuous differentiable functions.

Definition II.1: Let $f \colon \Omega \to \mathbb{R}$ be a function. If f belongs to $C^2(\Omega)$ and solves *Laplace's equation* $\partial_1^2 f + \dots + \partial_N^2 f \equiv 0$ (i.e. $\Delta f \equiv 0$, where again $\Delta := \sum_{i=1}^n \partial_i^2 \colon C^2(\Omega) \to C(\Omega)$ denotes the *Laplace operator*), f is called *harmonic*. We denote by $H(\Omega)$ the set of all harmonic functions on Ω.

Remark II.2: (i) Since the Laplace operator $\Delta \colon C^2(\Omega) \to C(\Omega)$ is linear and $H = \ker(\Delta \colon C^2(\Omega) \to C(\Omega))$, it follows that $H(\Omega)$ is a vector space. It contains all (locally) affine linear functions (i.e. functions f for which there are $a \in \mathbb{R}^N$ and $b \in \mathbb{R}$ such that for all $x \in \Omega$ it holds $f(x) = \langle x, a \rangle + b$). For $N = 1$, these are clearly all harmonic functions, thus we only consider $N \geq 2$ in the following.

(ii) If $\phi \colon \mathbb{R}^N \to \mathbb{R}^N$ is an isometry (i.e. there are $Q \in O(N)$ and $a \in \mathbb{R}^N$ such that for all $x \in \Omega$ it holds $\phi(x) = a + Qx$) or an dilation (i.e. there is some positive real number α such that for all $x \in \Omega$ we have $\phi(x) = \alpha x$), then for all $f \in H(\Omega)$, the composition $f \circ \phi$ belongs to $H(\phi(\Omega))$.

(iii) If $\Omega' \subseteq \Omega$ is an open, non-empty subset, then clearly for any $f \in H(\Omega)$, the restriction $f|_{\Omega'}$ belongs to $H(\Omega')$.

Theorem II.3: *Let $y \in \mathbb{R}^N$ be given. Then $U_y \colon \mathbb{R}^N - \{y\} \to \mathbb{R}$ defined via*

$$U_y(x) = \begin{cases} -\log\|x-y\|, & \text{if } N = 2, \\ \|x-y\|^{2-N}, & \text{if } N \geq 3, \end{cases}$$

where $x \in \mathbb{R}^N - \{y\}$, is harmonic on $\mathbb{R}^N - \{y\}$. Moreover, if f is harmonic on some annular region $A(y; r_1, r_2) := \{x \in \mathbb{R}^N \mid r_1 < \|x-y\| < r_2\}$, with $0 \leq r_1 < r_2 \leq \infty$ and depends only on $\|x-y\|$, then there are real numbers α and β such that $f = \alpha U_y + \beta$.

Proof: Suppose that $f \in C^2(A(y; r_1, r_2))$ depends only on $\|x-y\|$, i.e. there exists a function $F \in C^2((r_1, r_2))$ such that for all $x \in A(y; r_1, r_2)$ it holds $f(x) = F(\|x-y\|)$. Put $r := \|x-y\|$. Then $\partial_i r = r^{-1}(x_i - y_i)$, so that $\partial_i f(x) = r^{-1}F'(r)(x_i - y_i)$ for $1 \leq i \leq N$ and

$$\partial_i^2 f(x) = F''(r)\left(\frac{x_i - y_i}{r}\right)^2 + F'(r)\left(\frac{1}{r} - \frac{(x_i - y_i)^2}{r^3}\right).$$

Thus $\Delta f(x) = F''(r) + r^{-1}(N-1)F'(r)$. Hence, f is harmonic on $A(y; r_1, r_2)$ if and only if F solves the ordinary differential equation

$$F''(r) + \frac{(N-1)}{r}F'(r) = 0$$

on the interval (r_1, r_2). The only solutions are of the form

$$F(r) = \begin{cases} -\alpha\log(r) + \beta, & \text{if } N = 2, \\ \alpha r^{2-N} + \beta, & \text{if } N \geq 3, \end{cases}$$

where $r \in (r_1, r_2)$. Thus, both assertions of the theorem are immediate. □

Definition II.4: We call the function U_y as defined in Theorem II.3 the *fundamental harmonic function for $\mathbb{R}^N$ with pole at y.*

Potential theory is the study of harmonic functions in the same way as function theory is the study of holomorphic functions.

Remark II.5: The set of harmonic functions $H(\Omega)$ on some open set Ω is a vector space but no algebra (with respect to the pointwise multiplication). In fact, one has for two functions $f, g \in H(\Omega)$ that their pointwise product $f \cdot g$ is harmonic on Ω is and only if $\langle \operatorname{grad} f, \operatorname{grad} g \rangle \equiv 0$ on Ω. Recall that $\operatorname{grad} f = (\partial_1 f, \ldots, \partial_N f)$ for every $f \in C^1(\Omega)$.

In the case $N = 2$, one observes a close relationship between potential theory and function theory. We identify $\mathbb{C} \cong \mathbb{R}^2$ as abelian groups in the usual way, i.e. a complex number $z = x + \mathrm{i}y$ is identified with the tuple (x, y).

Let $\varnothing \neq \Omega \subseteq \mathbb{C}$ be open. We say that $f\colon \Omega \to \mathbb{C}$ is *holomorphic on* Ω if, for all $z_0 \in \Omega$, the limit

$$f'(z_0) := \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

exists. We denote by $\mathcal{O}(\Omega)$ the space of all holomorphic functions $f\colon \Omega \to \mathbb{C}$. For $f \in \mathcal{O}(\Omega)$, in particular the partial derivatives do exist since for a point $z_0 = x_0 + \mathrm{i}y_0 \in \Omega$ we have

$$\frac{\partial f}{\partial x}(z_0) = \lim_{x \to x_0} \frac{f(x + \mathrm{i}y_0) - f(x_0 + \mathrm{i}y_0)}{x - x_0} = f'(z_0)$$

$$\frac{\partial f}{\partial y}(z_0) = \lim_{y - y_0} \frac{f(x_0 + \mathrm{i}y) - f(x_0 + \mathrm{i}y_0)}{y - y_0} = \mathrm{i} f'(z_0). \quad \text{(II.1)}$$

For a partially differentiable function $f\colon \Omega \to \mathbb{C}$, we define the *Pompeiu-Wirtinger derivatives* by

$$\frac{\partial f}{\partial z}(z_0) := \frac{1}{2}\left(\frac{\partial f}{\partial x}(z_0) - \mathrm{i}\frac{\partial f}{\partial y}(z_0)\right), \qquad \frac{\partial f}{\partial \bar{z}}(z_0) := \frac{1}{2}\left(\frac{\partial f}{\partial x}(z_0) + \mathrm{i}\frac{\partial f}{\partial y}(z_0)\right).$$

For $f \in \mathcal{O}(\Omega)$, we see that for all $z_0 \in \Omega$ it holds $\frac{\partial f}{\partial z}(z_0) = f'(z_0)$ and $\frac{\partial f}{\partial \bar{z}}(z_0) = 0$. Function theory teaches us that the holomorphic functions $\mathcal{O}(\Omega)$ on Ω are contained in the smooth functions $C^\infty(\Omega, \mathbb{C})$ on Ω. Furthermore, a function $f\colon \Omega \to \mathbb{C}$ is holomorphic if and only if f belongs to $C^1(\Omega, \mathbb{C})$ and $\frac{\partial f}{\partial \bar{z}} \equiv 0$ on Ω.

For $\Delta\colon C^2(\Omega, \mathbb{C}) \to C(\Omega, \mathbb{C})$, where $f = u + \mathrm{i}v$ is sent to $\Delta u + \mathrm{i}\Delta v$, we find that for all $f \in C^2(\Omega, \mathbb{C})$ it holds

$$\Delta f = 4\frac{\partial^2 f}{\partial z \partial \bar{z}} = 4\frac{\partial^2 f}{\partial \bar{z} \partial z}. \quad \text{(II.2)}$$

Hence, for every holomorphic function $f\colon \Omega \to \mathbb{C}$, the Laplacian of f vanishes on the whole of Ω and thus $\operatorname{Re}(f)$ and $\operatorname{Im}(f)$ are both harmonic functions on Ω.

Theorem II.6: *Let $\varnothing \neq \Omega \subseteq \mathbb{C}$ be a simply connected domain. Then, for every $u \in H(\Omega)$, there is an $f \in \mathcal{O}(\Omega)$ such that $u = \mathrm{Re}(f)$. We call the imaginary part v of f, which is harmonic on Ω, the* harmonic conjugate of u.

Proof: Let u be harmonic on Ω. Then $h := 2\partial_z u \in C^1(\Omega, \mathbb{C})$ and, by Eq. (II.2), we find

$$0 \equiv \Delta u = 4\frac{\partial^2 u}{\partial \bar{z} \partial z} = 2\frac{\partial h}{\partial \bar{z}}.$$

Thus, h is holomorphic on $\mathcal{O}(\Omega)$. Since Ω is simply connected, there is a function $f_0 \in \mathcal{O}(\Omega)$ with $f_0' = h$. Using Eq. (II.1), we find that

$$\frac{\partial}{\partial x}(\mathrm{Re}(f_0) - u) = \mathrm{Re}\left(\frac{\partial f_0}{\partial x}\right) - \frac{\partial u}{\partial x} = \mathrm{Re}(f_0') - \mathrm{Re}(h) = 0,$$
$$\frac{\partial}{\partial y}(\mathrm{Re}(f_0) - u) = \mathrm{Re}\left(\frac{\partial f_0}{\partial y}\right) - \frac{\partial u}{\partial y} = -\mathrm{Im}(f_0') + \mathrm{Im}(h) = 0,$$

and hence we conclude that $\mathrm{Re}(f_0) - u \colon \Omega \to \mathbb{R}$ is constant, say $\mathrm{Re}(f_0) - u \equiv c$ for some $c \in \mathbb{R}$. Then, $f := f_0 - c$ does the job. □

Chapter III.

The Mean Value Property and its Consequences

Motivation III.1: Let $\varnothing \neq \Omega \subseteq \mathbb{C}$ be open and let f be holomorphic on Ω. For a point $z_0 \in \Omega$ and a radius $r > 0$ such that $\mathrm{cl}(D(z_0, r)) \subseteq \Omega$, we have

$$f(z_0) = \frac{1}{2\pi \mathrm{i}} \int_{\partial D(z_0,r)} \frac{f(z)}{z - z_0}\, dz,$$

where the boundary is parametrised via $\gamma\colon [0, 2\pi] \to \mathbb{C},\ t \mapsto z_0 + r\exp(\mathrm{i}t)$. Thus,

$$\begin{aligned} f(z_0) &= \frac{1}{2\pi \mathrm{i}} \int_0^{2\pi} \frac{f(\gamma(t))}{\gamma(t) - z_0} \gamma'(t)\, dt \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + r\exp(\mathrm{i}t))\, dt = \frac{1}{2\pi} \int_{\mathbb{S}^1} f(z_0 + r\zeta)\, d\sigma^1(\zeta), \end{aligned}$$

where $\sigma^1(\{\exp(\mathrm{i}t) \mid t \in (t_1, t_2)\}) = t_2 - t_1$, i.e. f has the *mean value property*. In particular,

$$f(z_0) \int_0^{r_0} r\, dr = \frac{1}{2} f(z_0) r_0^2 = \frac{1}{2\pi} \int_0^{r_0} \int_0^{2\pi} f(z_0 + r\exp(\mathrm{i}t)) r\, dt\, dr$$

and hence, $f(z_0) = (\pi r_0^2)^{-1} \int_{D(z_0,r_0)} f(z)\, d\lambda^2(z)$. This has many important consequences such as the famous *maximus modulus principle*. From Theorem II.6 it follows that every $u \in H(\Omega)$ has the mean value property. This is true not only in the case $N = 2$ but in full generality.

Definition III.2: Let $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ be open and let u be a continuous function on Ω. We say that u *has the mean value property on* Ω if for each $x_0 \in \Omega$ and

every radius $r > 0$ with $\mathrm{cl}(B(x_0, r)) \subseteq \Omega$, where $B(x_0, r)$ is the ball around x_0 with radius r with respect to the euclidean distance, it holds

$$u(x_0) = \frac{1}{N\omega_N} \int_{\mathbb{S}^{N-1}} u(x_0 + r\zeta)\, d\sigma^{N-1}(\zeta) =: \mathcal{M}(u; x_0, r),$$

or equivalently (see Exercise 2 (ii) on exercise sheet 1B)

$$u(x_0) = \frac{1}{\omega_N r^N} \int_{B(x_0,r)} u(x)\, d\lambda^N(x) =: \mathcal{A}(u; x_0, r).$$

Here λ^N is the Lebesgue measure on $\mathbb{R}^N$ and σ^{N-1} is the spherical measure on $\mathbb{S}^{N-1}$ defined via $\sigma^{N-1}(A) := N\lambda^N(\{ta \mid a \in A, t \in [0,1]\})$ for a Borel set $A \in \mathfrak{B}(\mathbb{S}^{N-1})$. For the spherical measure it holds

$$\int_{B(x_0,r_0)} f(x)\, d\lambda^N(x) = \int_0^{r_0} r^{N-1} \int_{\mathbb{S}^{N-1}} f(x_0 + r\zeta)\, d\sigma^{N-1}(\zeta)\, dr\,. \tag{III.1}$$

Finally, $\omega_N \lambda^N(B(0,1)) = \Gamma(N/2+1)^{-1}\pi^{N/2}$ is the volume of the N-sphere. Note that $\lambda^N(B(x_0, r)) = \omega_N r^N$ and $\sigma^{N-1}(\mathbb{S}^{N-1}) = N\omega_N$.

We want to prove the following more general statement:

Theorem III.3: *Let $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ be open. Then every $u \in H(\Omega)$ has the mean value property.*

The proof relies on the following fact:

Theorem III.4 (Gauss' Divergence Theorem): *If $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ is open and has a piecewise smooth boundary $\partial\Omega$, then every function $F = (f_1, \ldots, f_N)$ in $(C(\mathrm{cl}(\Omega)) \cap C^1(\Omega))^N$ satisfies*

$$\int_\Omega (\operatorname{div} F)(x)\, d\lambda^N(x) = \int_{\partial\Omega} \langle F(x), \nu(x)\rangle\, d\sigma_{\partial\Omega}(x)\,.$$

Here, $\nu\colon \partial\Omega \to \mathbb{R}^N$ are the outer unit normal vectors to the surface $\partial\Omega$, $\sigma_{\partial\Omega}$ is the surface measure on $\partial\Omega$ and $\operatorname{div} F$ is the divergence of R, which is defined by $\operatorname{div} F(x) := \sum_{i=1}^N \partial_i F^i(x)$.

Proof (of Theorem III.3): We apply Theorem III.4 to $B(x_0, r)$ and $\operatorname{grad} u$. We notice that $\nu(x) = r^{-1}(x - x_0)$ and thus we get for $u \in C^2(\Omega)$ that

$$\int_{B(x_0,r)} \Delta u(x)\, d\lambda^N(x) = \int_{B(x_0,r)} (\operatorname{div}\operatorname{grad} u)(x)\, d\lambda^N(x)$$

$$= \int_{\partial B(x_0,r)} \langle \operatorname{grad} u(x), r^{-1}(x-x_0)\rangle \, d\sigma_{\partial B(x_0,r)}(x)\,.$$

Additionally note that $\int_{\partial B(x_0,r)} f(x)\, d\sigma_{\partial B(x_0,r)} = r^{N-1}\int_{\mathbb{S}^{N-1}} f(x_0+r\zeta)\, d\sigma^{N-1}(\zeta)$. Using this fact, we can rephrase

$$\begin{aligned}\int_{B(x_0,r)} (\Delta u)(x)\, d\lambda^N(x) &= r^{N-1}\int_{\mathbb{S}^{N-1}} \langle \operatorname{grad} u(x_0 - r\zeta), \zeta\rangle \, d\sigma^{N-1}(\zeta)\\ &= r^{N-1}\int_{\mathbb{S}^{N-1}} \frac{\partial u}{\partial r}(x_0+r\zeta)\, d\sigma^{N-1}(\zeta)\\ &= r^{N-1}\frac{\partial}{\partial r}\int_{\mathbb{S}^{N-1}} u(x_0+r\zeta)\, d\sigma^{N-1}(\zeta),\end{aligned}$$

which yields the forumla

$$r\mathcal{A}(\Delta u; x_0, r) = N\frac{d}{dr}\mathcal{M}(u; x_0, r). \tag{III.2}$$

Thus, if $\Delta u \equiv 0$, then $\mathcal{M}(u; x_0, \cdot)\colon (0, r_0) \to \mathbb{R}$ must be constant, where $r_0 > 0$ is such that $B(x_0, r_0) \subseteq \Omega$. By part (i) of exercise 2 on exercise sheet 1B, it holds $\lim_{r\downarrow 0}\mathcal{M}(u; x_0, r) = u(x_0)$, hence for all $r \in (0, r_0)$ it holds $\mathcal{M}(u; x_0, r) = u(x_0)$. □

We will see that the local mean value property characterises harmonic functions. The following is the first step

Theorem III.5: *Let $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ be open. Suppose that $u \in C(\Omega)$ has the mean value property. Then u is smooth on Ω.*

Proof: We consider a smooth function ϕ on $\mathbb{R}$ such that $\phi(t) = 0$ for all $t \leq 0$ and

$$N\omega_M \int_0^1 t^{N-1}\phi(1-t^2)\, dt = 1.$$

For each natural number N, $\phi_n(x) := n^N\phi(1 - n^2\|x\|^2)$ defines a smooth function on $\mathbb{R}^N$. Further, we put

$$\Omega_n := \begin{cases}\{x \in \Omega \mid \operatorname{dist}(x, \partial\Omega) > 1/n\}, & \text{if } \Omega \neq \mathbb{R}^N,\\ \mathbb{R}^N, & \text{if } \Omega = \mathbb{R}^N\end{cases}.$$

Since ϕ_n and all its derivations are supported on $\operatorname{cl}(B(0, 1/n))$, we see that

$$u_n(x) := \int_\Omega \phi_n(x-y)u(y)\, d\lambda^n(y)$$

declares a smooth function $u_n\colon \Omega_n \to \mathbb{R}$. Now, for $x \in \Omega_n$, we have

$$\begin{aligned} u_n(x) &= \int_{B(x,1/n)} \phi_n(x-y)u(y)\,d\lambda^N(y) \\ &= \int_0^{1/n} r^{N-1} \int_{\mathbb{S}^{N-1}} \phi_n(r\zeta)u(x+r\zeta)\,d\sigma^{N-1}(\zeta)\,dr \\ &= \int_0^{1/n} r^{N-1}n^N\phi(1-u^2r^2) \int_{\mathbb{S}^{N-1}} u(x+r\zeta)\,d\sigma^{N-1}(\zeta) \\ &= u(x)N\omega_N \int_0^{1/n} r^{N-1}n^N\phi(1-u^2r^2)\,dr = u(x). \end{aligned}$$

This tells us that $u|_{\Omega_n} = u_n$ is smooth. Since $\Omega = \bigcup_{n=1}^{\infty} \Omega$, it follows that u is a smooth function on Ω. □

Theorem III.6: *Let $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ be open and consider $u \in C(\Omega)$. Then, the following are equivalent:*

(i) *u is harmonic on Ω,*

(ii) *u has the mean value property on Ω,*

(iii) *For all $x_0 \in \Omega$ there is some $r_0 > 0$ such that $B(x_0, r_0) \subseteq \Omega$ and $u|_{B(x_0,r)}$ has the mean value property on $B(x_0, r)$.*

Proof: The implication "(i) ⇒ (ii)" is just Theorem III.3 and the implication "(ii) ⇒ (iii)" is trivial.

For "(iii) ⇒ (i)", we note that due to Theorem III.5, $u|_{B(x_0,r_0)}$ is smooth. It suffices to show that $\Delta u(x_0) = 0$. From Eq. (III.1) and part (i) of exercise 2 of exercise sheet 1B, we infer that for every $r \in (0, r_0)$ it holds

$$N(\mathcal{M}(u; x_0, r) - u(x_0)) = \int_0^r \rho\mathcal{A}(\Delta u; x_0, \rho)\,d\rho,$$

and thus, again with part (i) of exercise 2, we obtain

$$N \lim_{r\downarrow 0} \frac{1}{r^2}(\mathcal{M}(u; x_0, r) - u(x_0)) = \lim_{r\downarrow 0} \frac{1}{r^2} \int_0^r \rho\mathcal{A}(\Delta u; x_0, \rho)\,d\rho = \frac{1}{2}\Delta u(x_0).$$

The mean value property now enforces $\Delta u(x_0) = 0$. □

Corollary III.7: *For every open subset $\varnothing \neq \Omega \subseteq \mathbb{R}^N$, we have the inclusion $H(\Omega) \subseteq C^\infty(\Omega)$. Moreover, if u is harmonic on Ω, then all partial derivatives of u belong to $H(\Omega)$.*

Proof: Take any $u \in H(\Omega)$. By Theorem III.3, u has the mean value property on Ω and thus, Theorem III.5 yields that $u \in C^\infty(\Omega)$. The additional assertion follows by induction by using that $\Delta \partial_k u = \partial_k \Delta u$ for all $u \in C^\infty(\Omega)$. $\square$

The following result is a "harmonic counterpart" of the maximum modulus principle for holomorphic functions.

Theorem III.8: *Let $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ be open and consider a holomorphic function u on Ω.*

(i) *If u attains a local maximum at some point $x_0 \in \Omega$ (i.e. there is $r_0 > 0$ such that $B(x_0, r_0) \subseteq \Omega$ and for all $x \in B(x_0, r_0)$ it holds $u(x) \leq u(x_0)$), then u is constant in a neighbourhood of x_0. In fact, u is constant on $B(x_0, r_0)$. The same condition holds for local minima.*

(ii) *If Ω is connected and u attains a local extremum at some point $x_0 \in \Omega$, then u is constant on Ω.*

(iii) *Denote by $\partial^\infty(\Omega)$ be the boundary of Ω in the one-point compactification $\mathbb{R}^N \cup \{\infty\}$. Note that $\infty \in \partial^\infty \Omega$ if and only if Ω is unbounded. If u is harmonic on Ω, then it holds*

$$\min\{u(x) \mid x \in \partial^\infty \Omega\} = \min\{u(x) \mid x \in \Omega \cup \partial^\infty \Omega\}, \tag{III.3}$$

$$\max\{u(x) \mid x \in \partial^\infty \Omega\} = \max\{u(x) \mid x \in \Omega \cup \partial^\infty \Omega\}. \tag{III.4}$$

Proof: (i) Since u has the mean value property, we see that for all $0 < r < r_0$ it holds

$$u(x_0) = \mathcal{A}(u; x_0, r) = \frac{1}{\omega_N N} \int_{B(x_0, r)} u(x)\, d\lambda^N(x) \leq u(x_0)$$

where by continuity of u, the last inequality is strict if

$$\{x \in B(x_0, r) \mid u(x) = u(x_0)\} \subsetneq B(x_0, r).$$

Thus u is constant on $B(x_0, r)$ for each $0 < r < r_0$ and so on $B(x_0, r_0)$.

(ii) Consider the set $\Omega_0 := \{x \in \Omega \mid u(x) = u(x_0)\}$. Note that Ω_0 is non-empty, as $x_0 \in \Omega_0$. By continuity of u, Ω_0 is closed relative to Ω. Due to (i), Ω_0 is also open. Hence, as Ω is connected, if follows that $\Omega_0 = \Omega$ and thus u is constant on the whole of Ω.

(iii) We prove Eq. (III.4), the proof of Eq. (III.3) is analogous. It suffices to show the inequality "$\geq$", the inequality "$\leq$" is obvious. Take $x_0 \in \Omega \cup \partial^\infty \Omega$ such that $u(x_0) = \max\{u(x) \mid x \in \Omega \cup \partial^\infty \Omega\}$ and let Ω_0 be any connected

component of Ω for which $x_0 \in \Omega_0 \cup \partial^\infty\Omega$. Now we have to distinguish cases: Firstly, if $x_0 \in \Omega_0$, by (ii) it follows that u is constant on Ω_0 and thus, by continuity of u, also on $\Omega_0 \cup \partial^\infty\Omega_0$. Hence

$$\begin{aligned}\max\{u(x) \mid x \in \partial^\infty\Omega\} &\geq \max\{u(x) \mid x \in \partial^\infty\Omega\} \\ &= u(x_0) = \max\{u(x) \mid x \in \Omega \cup \partial^\infty\Omega\}.\end{aligned}$$

Secondly, if $x_0 \in \partial^\infty\Omega_0$, then

$$\begin{aligned}\max\{u(x) \mid x \in \partial^\infty\Omega\} &\geq \max\{u(x) \mid x \in \partial^\infty\Omega\} \\ &\geq u(x_0) = \max\{u(x) \mid x \in \Omega \cup \partial^\infty\Omega\},\end{aligned}$$

which closes the proof. □

Remark III.9: Note that the proof of Theorem III.8 exclusively relies on the mean value property of u. Due to Theorem III.6 this is however equivalent to u being harmonic. In fact, only the following condition is needed: For all $x_0 \in \Omega$ is some radius $r_0 = r_0(x) > 0$ such that $B(x_0, r_0)$ is contained in Ω and for all $0 < r < r_0$ it holds

$$u(x) = \mathcal{A}(u; x_0, r) \quad \text{or equivalently} \quad u(x) = \mathcal{M}(u; x_0, r). \tag{III.5}$$

This seems to be a weaker assumption, but in Chapter 4, we will prove the following result, which says that even Eq. (III.5) is equivalent to u being harmonic.

Theorem III.10: *Let $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ be open and consider $u \in C(\Omega)$. Then, the following are equivalent:*

(i) *u is harmonic on Ω,*

(ii) *u has the (local) mean value property,*

(iii) *u satisfies condition Eq.* (III.5)*.*

The following is an analogue of Liouville's Theorem for holomorphic functions.

Theorem III.11: *Let u be harmonic on $\mathbb{R}^N$ and bounded from below (or from above). Then u is constant.*

Proof: Without loss of generality, we may suppose that for all $x \in \mathbb{R}^N$ it holds that $u(x) \geq 0$. Take $x, y \in \mathbb{R}^N$ and put $d := \|x-y\|$. Then, for every $r > 0$, the ball $B(x, r)$ is contained in $B(y, r + d)$ and hence by the mean value property and the positivity of u, we have

$$\begin{aligned} u(x) &= \mathcal{A}(u; x, r) \\ &\leq \frac{1}{\omega_N r^N} \int_{B(y,r+d)} u(x)\, d\lambda^N(x) \\ &= \left(\frac{r+d}{r}\right)^N \mathcal{A}(u; y, r+d) = \left(1 + \frac{d}{r}\right)^N u(y). \end{aligned}$$

If r gets arbitrarily large, the right hand side of the above equation tends to $u(y)$. This shows that $u(x) \leq u(y)$. Since x and y were arbitrary, it follows that u is constant. □

Chapter IV.

The Poisson Integral Formula for a Ball

The Poisson integral formula for balls can be seen as a "harmonic analogue" of Cauchy's integral formula for holomorphic funtions on disks, see Exercise 2 on exercise sheet 2B.

Definition IV.1 (Poisson kernel): Let $x_0 \in \mathbb{R}^N$ and $r > 0$ be given. The function

$$K_{x_0,r}\colon B(x_0,r) \times \partial B(x_0,r) \longrightarrow \mathbb{R}, \qquad K_{x_0,r}(x,y) := \frac{1}{N\omega_N r} \frac{r^2 - \|x-x_0\|^2}{\|x-y\|^N}$$

is called the *Poisson kernel of* $B(x_0,r)$.

Lemma IV.2: *Let* $x_0 \in \mathbb{R}^N$ *and* $r > 0$ *be given. For every* $y \in \partial B(x_0,r)$*, the function* $B(x_0,r) \to \mathbb{R}, x \mapsto K_{x_0,r}(x,y)$ *is harmonic on* $B(x_0,r)$*.*

This assertion will be left as an exercise for the reader – namely in Exercise 1 on exercise sheet 2B.

Definition IV.3 (Poisson integral of a signed measure): Let $x_0 \in \mathbb{R}^N$ and let $r > 0$ be given. For a signed measure $\mu\colon \mathfrak{B}(\partial B(x_0,r)) \to \mathbb{R}$ (i.e. a σ-additive function on the σ-algebra of Borel sets on $\partial B(x_0,r)$ with $\mu(\varnothing) = 0$ that does not take values $\pm\infty$), we call the function

$$I_{\mu,x_0,r}\colon B(x_0,r) \longrightarrow \mathbb{R}, \qquad I_{\mu,x_0,r}(x) := \int_{\partial B(x_0,r)} K_{x_0,r}(x,y)\, d\mu(y)\,.$$

the *Poisson integral of* μ.

A better intution for signed measures is instilled by the theorem on the so-called Hahn-Jordan decomposition:

Theorem: *If X is a topological space and $\mu\colon \mathfrak{B}(X) \to \mathbb{R}$, then there are finite measures $\mu^{\pm}$ and disjoint $P, N \in \mathfrak{B}(X)$ with $P \cup N = X$ such that $\mu^+(N) = 0 = \mu^-(P)$ and $\mu = \mu^+ - \mu^-$.*

In this situation, $\|\mu\| := \mu^+(X) + \mu^-(X)$ is called *total varitation of* μ.

If $f\colon \partial B(x_0, r) \to \mathbb{R}$ is integrable with respect to the surface measure $\sigma_{\partial B(x_0,r)}$, then we put $I_{f,x_0,r} := I_{\mu,x_0,r}$, where μ is the signed measure given by

$$d\mu(x) = f(y)\, d\sigma_{\partial B(x_0,r)}(y)\,.$$

Theorem IV.4: *Let $x_0 \in \mathbb{R}^N$ and $r > 0$ be given.*

(i) *If μ is a signed measure on $\partial B(x_0, r)$, then $I_{\mu,x_0,r}$ is harmonic on $B(x_0, r)$.*

(ii) *If $f\colon \partial B(x_0, r) \to \mathbb{R}$ is integrable with respect to the surface measure $\sigma_{\partial B(x_0,r)}$, then for $y \in \partial B(x_0, r)$, we have*

$$\limsup_{B(x_0,r)\ni x\to y} I_{f,x_0,r}(x) \le \limsup_{\partial B(x_0,r)\ni z\to y} f(z). \tag{IV.1}$$

Furthermore, if f is continuous on $\partial B(x_0, r)$, then

$$\lim_{B(x_0,r)\ni x\to y} I_{f,x_0,r}(x) = f(y). \tag{IV.2}$$

Proof: (i) Take any $\mathrm{cl}(B(x, \rho)) \subseteq B(x_0, r)$. By Fubinis Theorem, we get

$$\begin{aligned}\mathcal{A}(I_{\mu,x_0,r}; x, \rho) &= \int_{\partial B(x_0,r)} \mathcal{A}(K_{x_0,r}(\cdot, y); x, \rho)\, d\mu(y)\\ &= \int_{\partial B(x_0,r)} K_{x_0,r}(x, y)\, d\mu(y) = I_{\mu,x_0,r}\end{aligned}$$

Hence, $I_{\mu,x_0,r}$ has the mean value property on $B(x_0, r)$. By Theorem III.6 we get that $I_{\mu,x_0,r}$ is harmonic on $B(x_0, r)$.

(ii) Firstly, for any constant $c \in \mathbb{R}$, we claim that $I_{c,x_0,r} \equiv c$. To see this, note that $I_{c,x_0,r}$ is harmonic on $B(x_0, r)$ and $I_{c,x_0,r}(x)$ depends only on $\|x - x_0\|$. Now, Theorem II.3 tells us that there are $\alpha, \beta \in \mathbb{R}$ such that for all $x \in B(x_0, r) - \{x_0\}$ it holds $I_{c,x_0,r} = \alpha U_{x_0} + \beta = A(x_0; 0, r)$. Since $\lim_{x\to x_0} I_{c,x_0,r}(x) = I_{c,x_0,r}(x_0) = c$, we must have $\alpha = 0$ and $\beta = c$ which yields the claim.

Secondly, suppose that for some $c \in \mathbb{R}$, it holds $\limsup_{\partial B(x_0,r)\ni z\to y} f(z) < c$. If no such c exists, i.e. if the limes superior in question is infinity, there is nothing to prove. If now there is such a c, there is some $\delta > 0$ such that

$$\forall\, z \in B(y, 2\delta) \cap \partial B(x_0, r) : f(z) < c. \tag{IV.3}$$

Then, for every $x \in B(y,\delta) \cap B(x_0, r)$, it holds

$$I_{f,x_0,r}(x) - c \equiv I_{f-c,x_0,r}(x) = h_1(x) - h_2(x),$$

where

$$h_1(x) = \int_{\partial B(x_0,r)-B(y,2\delta)} K_{x_0,r}(x,z)(f(z)-c)\, d\sigma_{\partial B(x_0,r)}(z),$$
$$h_2(x) = \int_{\partial B(x_0,r)\cap B(y,2\delta)} K_{x_0,r}(x,z)(f(z)-c)\, d\sigma_{\partial B(x_0,r)}(z);$$

by Eq. (IV.3), we have $h_2(x) < 0$ and further

$$\begin{aligned}
|h_1(x)| &\le \max_{z\in\partial B(x_0,r)-B(y,2\delta)} K_{x_0,r}(x,z) \int_{\partial B(x_0,r)} (|f(z)| + |c|)\, d\sigma_{\partial B(x_0,r)}(z) \\
&\le \frac{1}{N\omega_N r} \frac{r^2 - \|x-x_0\|^2}{\delta^N} \int_{\partial B(x_0,r)} (|f(z)| + |c|)\, d\sigma_{\partial B(x_0,r)}(z) \\
&\le \frac{1}{N\omega_N r} \frac{r^2 - \|x-x_0\|^2}{\delta^N} \xrightarrow{x\to y} 0.
\end{aligned}$$

Hence, $\limsup_{B(x_0,r)\ni x\to y} I_{f,x_0,r}(x) \le c$, which shows Eq. (IV.1).

(iii) For a continuous function $f\colon \partial B(x_0,r) \to \mathbb{R}$, we apply Eq. (IV.1) to f and $-f$, which yields the proof of Eq. (IV.2). □

Theorem IV.5 (Possions Integral Formula): *Let $x_0 \in \mathbb{R}^N$ and let $r > 0$ be given. Then, every function $u \in C(\mathrm{cl}(B(x_0,r))) \cap H(B(x_0,r))$ satisfies for all $x \in B(x_0,r)$ that*

$$u(x) = I_{u|_{\partial B(x_0,r)},x_0,r}(x).$$

Proof: By Theorem IV.4 (i), we have that $v := u - I_{u,x_0,r}$ is harmonic on $B(x_0,r)$ and due to Theorem IV.4 (ii), we have for all $y \in \partial B(x_0,r)$ that

$$\lim_{B(x_0,r)\ni x\to y} v(x) = 0.$$

Thus, v extends to a function $v \in H(B(x_0,r)) \cap C(\mathrm{cl}(B(x_0,r)))$ such that $v|_{\partial B(x_0,r)} \equiv 0$ and by Theorem III.8 (iii), it follows $v \equiv 0$, as desired. □

Proof (of Theorem III.10): The implication "(i) ⇒ (ii)" was shown in Theorem III.3 and the implication "(ii) ⇒ (iii)" is trivial.

For "(iii) ⇒ (i)", take any $x_0 \in \Omega$ and $r > 0$ such that $\mathrm{cl}(B(x_0, r)) \subseteq \Omega$. We consider the function $v := u - I_{u,x_0,r}$. By Theorem IV.4 (i), v is continuous on $B(x_0, r)$ and due to Theorem IV.4 (ii) we see that v extends to a function $v \in C(\mathrm{cl}(B(x_0, r)))$ with $v|_{\partial B(x_0,r)} \equiv 0$.

Since $I_{u,x_0,r}$ is harmonic on $B(x_0, r)$ and u satisfies Eq. (III.1) by assumption, also v satisfies Eq. (III.1). Thus, Theorem III.8 (iii) can be applied (see Remark III.9) which gives $v \equiv 0$ on $\mathrm{cl}(B(x_0, r))$ and thus the restriction of u to $B(x_0, r)$ equals $I_{u,x_0,r}$, which is harmonic on $B(x_0, r)$. In total, u is harmonic on Ω as desired. □

Theorem IV.6 (Harnack's Inequalitites): *Let $x_0 \in \mathbb{R}^N$ and let $r > 0$ be given. Suppose that u is harmonic on $B(x_0, r)$ satisfying $u(x) \geq 0$ for all $x \in B(x_0, r)$. Then,*

$$\frac{(r - \|x - x_0\|)r^{N-2}}{(r + \|x - x_0\|)^{N-1}} u(x_0) \leq u(x) \leq \frac{(r + \|x - x_0\|)r^{N+2}}{(r - \|x - x_0\|)^{N-1}} u(x_0)$$

holds for every point x in $B(x_0, r)$.

Proof: Take any $0 < \rho < r$. Then, by Theorem IV.5, we have for all $x \in B(x_0, \rho)$ that $u(x) = I_{u,x_0,\rho}(x)$. Note that, for every x in $B(x_0, \rho)$ and y in $\partial B(x_0, \rho)$,

$$\begin{aligned} K_{x_0,\rho}(x, y) &= \frac{1}{\omega_N N \rho} \frac{\rho^2 - \|x - x_0\|^2}{\|x - y\|^N} \\ &\geq \frac{1}{\omega_N N \rho} \frac{(\rho - \|x - x_0\|)(\rho + \|x - x_0\|)}{(\|y - x_0\| + \|x - x_0\|)^N} = \frac{1}{\omega_N N_\rho} \frac{\rho - \|x - x_0\|}{(\rho + \|x + x_0\|)^{N-1}}. \end{aligned}$$

Similarly, we deduce

$$K_{x_0,\rho}(x, y) \leq \frac{1}{\omega_N N \rho} \frac{\rho + \|x - x_0\|}{(\rho - \|x - x_0\|)^{N-1}}.$$

Hence, by the mean value property of u, we find

$$\begin{aligned} u(x) &= \int_{\partial B(x_0,r)} K_{x_0,\rho}(x, y) u(y) \, d\sigma_{\partial B(x_0,\rho)}(y) \\ &\geq \frac{1}{\omega_N N_\rho} \frac{\rho - \|x - x_0\|}{(\rho + \|x - x_0\|)^{N-1}} \int_{\partial B(x_0,\rho)} u(y) \, d\sigma_{\partial B(x_0,\rho)}(y) \end{aligned}$$

$$= \frac{1}{\omega_N N \rho} \frac{\rho - \|x - x_0\|}{(\rho + \|x - x_0\|)^{N-1}} \rho^{N-1} N \omega_N u(x_0)$$
$$= \frac{(\rho - \|x - x_0\|)\rho^{N-2}}{(\rho + \|x - x_0\|)^{N-1}} u(x_0).$$

Again, with the same arguments, we see

$$u(x) \leq \frac{(\rho + \|x - x_0\|)\rho^{N-2}}{(\rho - \|x - x_0\|)^{N-1}} u(x_0)$$

Letting $\rho \uparrow r$, we obtain the asserted bounds. □

Corollary IV.7: *In the situation of Theorem IV.6, it holds that*

$$\|\operatorname{grad} u(x_0)\| \leq \frac{N}{r} u(x_0).$$

Proof: Let $e \in \mathbb{R}^N$ with $\|e\| = 1$ be given. Then

$$f\colon (-r, r) \longrightarrow \mathbb{R}, \qquad t \longmapsto u(x_0 - te)$$

is well-defined and smooth as composition of smooth functions, whose derivative in zero is given by $f'(0) = \langle \operatorname{grad} u(x_0), e \rangle$. By Theorem IV.6, we further have for $t \in (0, r)$ that

$$\frac{(r-t)r^{N-2}}{(r+t)^{N-1}} f(0) \leq f(t) \leq \frac{(r+t)r^{N-2}}{(r-t)^{N-1}} f(0).$$

Hence,

$$\frac{1}{t}\left(\frac{(r-t)r^{N-2}}{(r+t)^{N-1}} - 1\right) f(0) \leq \frac{f(t) - f(0)}{t} \leq \frac{1}{t}\left(\frac{(r+t)r^{N-2}}{(r-t)^{N-1}} - 1\right) f(0).$$

Letting $t \downarrow 0$, we get the bounds $-N/r \leq f'(0) \leq N/r$. Therefore, we have the estimate $|\langle \operatorname{grad} u(x_0), e \rangle| = |f'(0)| \leq N/r$, from which the assertion follows. □

Remark IV.8: The *Dirichlet Problem on* $B(x_0, r)$ is to find, for a given function $f \in C(\partial B(x_0, r))$, a function u which is harmonic on $B(x_0, r)$ such that for all $y \in \partial B(x_0, r)$ it holds

$$\lim_{B(x_0,r) \ni x \to y} u(x) = f(y).$$

Theorem IV.4 tells us that $u = I_{f,x_0,r}$ solves the Dirichlet Problem. Due to Theorem III.8, this solution is in fact unique.

Chapter V.

Subharmonic functions

Subharmonic functions generalise harmonic functions; they are more flexible but have similar strong properties. This class is crucial for the study of harmonic functions.

Definition V.1: Let X be a topological space.

A function $f\colon X \to \mathbb{R} \cup \{-\infty\}$ is called *upper semicontinuous on* X or just *upper semicontinuous*, if $f^{-1}([-\infty, a))$ is open in X for any real number a.

A function $f\colon X \to \mathbb{R} \cup \{-\infty\}$ is called *lower semicontinuous on* X or just *lower semicontinuous*, if $-f$ is upper semicontinuous on X.

Definition V.2: Let $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ be open. A function $s\colon \Omega \to [-\infty, +\infty)$ is called *subharmonic on* Ω or briefly *subharmonic*, if s is upper semicontinuous on Ω, s has the *subharmonic mean value property on* Ω, i.e. for all $r > 0$ with $\mathrm{cl}(B(x,r)) \subseteq \Omega$ it holds $s(x) \leq \mathcal{M}(s; x, r)$ and $s \not\equiv -\infty$ on each connected component of Ω. By $S(\Omega)$, we denote the set of all subharmonic functions on Ω.

A function $u\colon \Omega \to (-\infty, +\infty]$ is called *superharmonic on* Ω or briefly *superharmonic*, if $-u\colon \Omega \to [-\infty, +\infty)$ is subharmonic. Those functions have the *superharmonic mean value property*, i.e. for all $r > 0$ with $\mathrm{cl}(B(x,r)) \subseteq \Omega$ it holds $u(x) \geq \mathcal{M}(u; x, r)$. We denote by $U(\Omega)$ the set of all superharmonic functions on Ω.

Note that the harmonic functions on Ω are precisely those that are both sub- and superharmonic on Ω.

Remark V.3: (i) Let $f\colon X \to [-\infty, +\infty)$ be upper semicontinuous and let $K \subseteq X$ be compact. Then $\sup\{f(x) \mid x \in K\} < \infty$ and there exists $x_0 \in K$ such that $f(x_0) = \sup\{f(x) \mid x \in K\}$.

(ii) Let $s\colon \Omega \to [-\infty, +\infty)$ be upper semicontinuous. Decompose s in its positive- and negative parts $s^+\colon \Omega \to [0, +\infty)$, $s^-\colon \Omega \to [0, +\infty]$ that are, for $x \in \Omega$, defined via $s^\pm(x) := \max\{\pm s(x), 0\}$. It holds $s = s^+ - s^-$ and s^+ is still upper semicontinuous. Thus, whenever $\mathrm{cl}(B(x,r)) \subseteq \Omega$, then

$$\begin{aligned}\mathcal{M}(s;x,r) &= \frac{1}{\omega_N N r^{N-1}} \int_{\partial B(x,r)} s(\zeta)\, d\sigma(\zeta) \\ &:= \frac{1}{\omega_N N r^{N-1}} \int_{\partial B(x,r)} s^+(\zeta)\, d\sigma(\zeta) - \frac{1}{\omega_N N r^{N-1}} \int_{\partial B(x,r)} s^-(\zeta)\, d\sigma(\zeta)\end{aligned}$$

where $\sigma = \sigma_{\partial B(x_0,r)}$. The integral over s^+ is bounded by $\max_{\zeta \in \partial B(x,r)} s^+(\zeta)$ which is finite and the integral of s^- certainly is something in $[0, \infty]$. Thus, $\mathcal{M}(s;x,r)$ is a value in $[-\infty, \infty)$ and hence well-defined.

Theorem V.4: *Let $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ be open and let $s \in S(\Omega)$. Then, we have:*

(i) *For all $y \in \Omega$, it holds* $\limsup_{x \to y} s(x) = s(y)$.

(ii) *For all $r > 0$ with* $\mathrm{cl}(B(x,r)) \subseteq \Omega$ *we have* $s(x) \leq A(s;x,r)$.

(iii) *The function s is* locally integrable*, i.e. for every compact subset $K \subseteq \Omega$ it holds*

$$\int_K |s(x)|\, d\lambda^N(x) < \infty.$$

Proof: (i) Since s is upper semicontinuous, we have $\limsup_{x \to y} s(x) \leq s(y)$. If this inequality were strict, then we could find $r_0 > 0$ such that $B(y, r_0) \subseteq \Omega$ and for all $x \in B(y, r_0) - \{y\}$ it holds $s(x) < s(y)$. Thus, for each $0 < r < r_0$, by Remark V.3(i),

$$\mathcal{M}(s;y,r) \leq \max\{s(\zeta) \mid \zeta \in \partial B(y,r)\} < s(y)$$

in contradiction to the subharmonic mean value property.

(ii) Like in Exercise 2 (ii) from sheet 1B, we get

$$r^N \mathcal{A}(s;x,r) = N \int_{[0,r]} \rho^N \mathcal{M}(s;x,\rho)\, d\lambda^1(\rho) \tag{V.1}$$

and derive that $\mathcal{A}(s;x,r) \geq s(x)$.

(iii) By Heine-Borel, it suffices to show that for each $x \in \Omega$, one finds $r > 0$ such that $\mathrm{cl}(B(x,r)) \subseteq \Omega$ and Eq. (V.1) holds for $K = \mathrm{cl}(B(x,r))$. In particular, withour loss of generality, we may suppose that Ω is connected. Put

$$\Omega_0 := \Big\{ x \in \Omega \;\Big|\; \exists\, r > 0 : \mathrm{cl}(B(x,r)) \subseteq \Omega, \int_{\mathrm{cl}(B(x,r))} |s(x)|\, d\lambda^N(x) < \infty \Big\}.$$

First of all, Ω_0 is open. If we take $x \in \Omega_0$, there is some radius $r > 0$ such that $\mathrm{cl}(B(x,r)) \subseteq \Omega$ and $\int_{\mathrm{cl}(B(x,r))} |s(x)|\, d\lambda^N(x) < \infty$. We want to show that the whole of $B(x,r)$ is contained in Ω_0. Take $x' \in B(x,r)$ and set $r' := r - |x' - x| > 0$. Then, $B(x',r') \subseteq B(x,r)$ and

$$\int_{\mathrm{cl}(B(x',r'))} |s(x)| d\lambda^N(x) \leq \int_{\mathrm{cl}(B(x,r)} |s(x)|\, d\lambda^N(x) < \infty,$$

i.e., $x' \in \Omega_0$, which shows $B(x,r) \subseteq \Omega_0$.

Secondly, $\Omega - \Omega_0$ is open and $s|_{\Omega-\Omega_0} \equiv -\infty$. To see this, let $x \in \Omega - \Omega_0$ be given. Choose $r > 0$ such that $\mathrm{cl}(B(x,2r)) \subseteq \Omega$. We want to show that $B(x,r)$ is contained in $\Omega - \Omega_0$ and that $s|_{B(x,r)} \equiv -\infty$. Let thus $x' \in B(x,r)$ and put $r' := r - |x' - x| \in (0, r]$. Then $B(x,r') \subseteq \Omega$ and hence, as $x \in \Omega - \Omega_0$, we must have $\int_{\mathrm{cl}(B(x,r'))} |s|\, d\lambda^N = \infty$. Since $\mathrm{cl}(B(x,r')) \subseteq \mathrm{cl}(B(x',r))$, we infer that also

$$\int_{\mathrm{cl}(B(x',r))} |s|\, d\lambda^N = \infty. \tag{V.2}$$

However, as $\mathrm{cl}(B(x',r)) \subseteq \mathrm{cl}(B(x,2r)) \subseteq \Omega$, we know that s is bounded from above on $\mathrm{cl}(B(x',r))$; hence, Eq. (V.2) gives that

$$\int_{\mathrm{cl}(B(x',r))} s\, d\lambda^N = -\infty$$

and so, $\mathcal{A}(s; x', r) = -\infty$. Due to part (ii), it follows that $s(x') = -\infty$. Therefore, $s|_{B(x,r)} \equiv -\infty$ and consequently $B(x,r) \subseteq \Omega - \Omega_0$.

Finally, by Definition 5.2 (iii), we know that $s \not\equiv -\infty$, thus $\Omega_0 \neq \varnothing$. In summary, as Ω is connected, we get $\Omega = \Omega_0$. □

Corollary V.5: *Let $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ be open. For each $s \in S(\Omega)$, we have $\lambda^N(\{x \in \Omega \mid s(x) = -\infty\}) = 0$.*

Proof: Let $(K_n)_{n\in\mathbb{N}}$ be a sequence of compact subsets K_n of Ω such that $\Omega = \bigcup_{n=1}^{\infty} K_n$. Put $E := \{x \in \Omega \mid s(x) = -\infty\}$. Then $\lambda^N(K_n \cap E) = 0$ since $\int_{K_n} |s(x)|\, d\lambda^N(x) < \infty$ due to Theorem V.4. But because $E = \bigcup_{n=1}^{\infty}(K_n \cap E)$, E is itself a null set. □

Example V.6: (i) If h is harmonic, then both $|h|$ and h^2 are subharmonic. This can be shown with help of the triangular inequality for integrals and the Cauchy Schwarz inequality.

(ii) It is a less obvious fact that $s\colon \Omega \to \mathbb{R},\ x \mapsto -\log(\operatorname{dist}(x, \partial\Omega))$ is subharmonic for every open set $\varnothing \neq \Omega \subsetneq \mathbb{R}^N$ where

$$\operatorname{dist}(x, \partial\Omega) = \inf\{|x - y| \mid y \in \partial\Omega\}.$$

is the distance of x to the set Ω.

The maximum principle for harmonic functions Theorem III.8 generalises to subharmonic functions.

Theorem V.7: *Let $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ be open and consider $s \in S(\Omega)$.*

(i) *If s attains a local maximum at some point $x_0 \in \Omega$, then s is constant on some neighbourhood of x_0.*

(ii) *If Ω is connected and s attains a local maximum at some point $x_0 \in \Omega$, then s is constant on Ω.*

(iii) *If $u \in U(\Omega)$ is such that for all $y \in \partial^\infty\Omega$ it holds*

$$\limsup_{x \to y}(s(x) - u(x)) \leq 0, \tag{V.3}$$

then for all $x \in \Omega$ it holds $s(x) \leq u(x)$.

Proof: Statements (i) and (ii) can be shown like in the proof of Theorem III.8. To prove statement (iii), we may suppose that $u \equiv 0$, since $s - u$ is subharmonic on Ω, and that Ω is connected, since Eq. (V.3) holds for each connected component of Ω.

We define an upper semicontinuous function $\bar{s}\colon \Omega \cup \partial^\infty\Omega \to [-\infty, +\infty)$, by $\bar{s}|_\Omega = s$ and for all $y \in \partial^\infty\Omega$ by $\bar{s}(y) := \limsup_{x \to y} s(x)$. By (Remark 5.3)(i), we find $x_0 \in \Omega \cup \partial^\infty\Omega$ such that $\bar{s}(x_0) = \sup\{\bar{s}(x) \mid x \in \Omega \cup \partial^\infty\Omega\}$. If we had $x_0 \in \Omega$, then s would have a constant positive value on Ω due to (ii). By Eq. (V.3) this value must be less or equal to zero. Otherwise, if $x_0 \in \partial^\infty\Omega$, then, for each $x \in \Omega$, $s(x) \leq \bar{s}(x_0) \leq 0$ by Eq. (V.3). □

Analogously, one has a minimum principle for superharmonic functions. Our next goal in the following is the characterisation of subharmonicity, see (Theorem 3.6), (Theorem 3.10) and (Theorem 4.5).

Theorem V.8: *Let $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ be open and consider an upper semicontinuous function $s\colon \Omega \to [-\infty, +\infty)$ which satisfies $s \not\equiv -\infty$ on each connected component of Ω. Then, the following are equivalent:*

(i) *The function s is superharmonic on Ω,*

(ii) *The function s is bounded from above by $I_{s,x,r}$ on $B(x,r)$ whenever* $\mathrm{cl}(B(x,r)) \subseteq \Omega$,

(iii) *For each $x \in \Omega$ such that $s(x) > -\infty$, we have*

$$\limsup_{r \downarrow 0} \frac{1}{r^2} \mathcal{M}(s;x,r) - s(x) \geq 0.$$

(iv) *For all $x \in \Omega$ there is some radius $r_0 > 0$ such that $B(x,r_0) \subseteq \Omega$ and for all $0 < r < r_0$ it holds $s(x) \leq \mathcal{M}(s;x,r)$.*

(v) *For all $x \in \Omega$ there is some radius $r_0 > 0$ such that $B(x,r_0) \subseteq \Omega$ and for all $0 < r < r_0$ it holds $s(x) \leq \mathcal{A}(s;x,r)$.*

(vi) *If U is an open and bounded set such that $\mathrm{cl}(U) \subseteq \Omega$ and if $h \in C(\mathrm{cl}(U))$ is harmonic on U and satisfies $s \leq h$ on ∂U, then $s \leq h$ on U.*

The proof requires the following fact:

Lemma V.9: *Let $\varnothing \neq X \subseteq \mathbb{R}^N$ be any subset and suppose $f\colon X \to [-\infty, +\infty)$ is upper semicontinuous on X and bounded from above. Then there is a pointwise decreasing sequence $(f_n)_{n\in\mathbb{N}}$ in $C(\mathbb{R}^N)$ such that for all $x \in X$ it holds $f_n(x) \to f(x)$.*

Proof: We will give a sketch of the proof of this statetement. We extend f to an upper semicontinuous function

$$\bar{f}\colon \mathbb{R}^N \longrightarrow [-\infty,+\infty), \qquad x \longmapsto \begin{cases} f(x), & \text{if } x \in X, \\ \limsup_{y\to x} f(y), & \text{if } x \in \mathrm{cl}(X) - X, \\ -\infty, & \text{else.} \end{cases}$$

If $\bar{f} \equiv -\infty$ on $\mathbb{R}^N$, then the sequence defined via $f_n \equiv -n$ does the trick. Otherwise, we define $f_n\colon \mathbb{R}^N \to \mathbb{R}$ via $f_n(x) := \sup_{y\in\mathbb{R}^N}(\bar{f}(y) - n\|x-y\|)$ and check that for all $x_1, x_2 \in \mathbb{R}^N$ it holds $|f_n(x_1) - f_n(x_2)| \leq n\|x_1 - x_2\|$. Thus, in this case, f_n is continuous on $\mathbb{R}^N$ and one verifies that f_n is pointwise decreasing and convergent to $\bar{f}$. □

Proof (of Theorem V.8): The implications "(i) ⇒ (iv)" and "(iv) ⇒ (iii)" are obvious.

For "(iii) ⇒(vi)" we consider the function $f\colon \mathbb{R}^N \to \mathbb{R}$, $y \mapsto \|y\|^2$ and put $a := \sup\{f(y) \mid y \in U\} < \infty$. For $\varepsilon > 0$ we put $u_\varepsilon := h - s - \varepsilon(f - a)$, which is a lower semicontinuous function on $\mathrm{cl}(U)$ satisfying $u_\varepsilon \geq 0$ on ∂U. Now we set $b_\varepsilon := \inf\{u_\varepsilon(y) \mid y \in \mathrm{cl}(U)\}$.

It holds $u_\varepsilon > b_\varepsilon$ on U, which can be seen as follows: Like in the proof of (Theorem III.6), we infer from (Eq. III.2)

$$\lim_{r\downarrow 0} \frac{1}{r^2}(\mathcal{M}(f;y,r) - f(y)) = \frac{1}{2N}(\Delta f)(y) = 1$$

for all $y \in \mathbb{R}^N$, and thus, for every $y \in U$, it holds

$$\begin{aligned}\mathcal{M}(u_\varepsilon;y,r) &= \mathcal{M}(h;y,r) - \mathcal{M}(s;y,r) - \varepsilon(\mathcal{M}(f;y,r) - a)\\ &= h(y)\mathcal{M}(s;y,r) - \varepsilon(\mathcal{M}(f;y,r) - a),\end{aligned}$$

and hence, for all sufficiently small $r > 0$, by (iii), we find

$$\begin{aligned}&\frac{1}{r^2}(\mathcal{M}(u_\varepsilon;y,r) - u_\varepsilon(y))\\ &\qquad = -\frac{1}{r^2}(\mathcal{M}(s;y,r) - s(y)) - \varepsilon\frac{1}{r^2}(\mathcal{M}(f;y,r) - f(y)) < 0,\end{aligned}$$

i.e. $\mathcal{M}(u_\varepsilon;y,r) < u_\varepsilon(y)$. Therefore, if there was a point $y_0 \in U$ such that $u_\varepsilon(y_0) \le b_\varepsilon$, then $b_\varepsilon \le \inf\{u_\varepsilon(y) \mid y \in B(y_0,r)\} \le \mathcal{M}(u_\varepsilon;y_0,r) < u_\varepsilon(y) \le b_\varepsilon$, which is a contradiction. Hence, $u_\varepsilon > b_\varepsilon$ on all of U.

Due to (Remark V.3)(i), we find $y_0 \in \mathrm{cl}(U)$ such that $u_\varepsilon(y_0) = b_\varepsilon$. By the preceeding claim, we must have $y_0 \in \partial U$ and hence $b_\varepsilon = u_\varepsilon(y_0) \ge 0$. Thus, $u_\varepsilon(y) \ge 0$ for all $y \in \mathrm{cl}(U)$.

Letting $\varepsilon \downarrow 0$, we infer from the non-negativity of u_ε on $\mathrm{cl}(U)$ that $h \ge s$ on all of U.

"(iv) $\Rightarrow$ (ii)": Consider $\mathrm{cl}(B(x,r)) \subseteq \Omega$. By Lemma V.9, there is a pointwise decreasing sequence $(f_n)_{n\in\mathbb{N}}$ in $C(\partial B(x,r))$ such that $f_n \to s$ on $\partial B(x,r)$. We define now define functions h_n that are continuous on $\mathrm{cl}(B(x,r))$ and harmonic on $B(x,r)$ via $b_n|_{B(x,r)} := I_{f_n,x,r}$ and $h_n|_{\partial B(x,r)} := f_n$. For the well-definedness of this sequence, check (Theorem IV.4)(ii). By assumption (vi), we have the estimate $s \le h_n$ on $B(x,r)$. Now, the Beppo-Levi-Theorem implies that $h_n \to I_{s,x,r}$ on $B(x,r)$. Therefore, $s \le I_{s,x,r}$ as desired.

For "(ii) $\Rightarrow$ (i)": If we take $B(x,r)$ such that $\mathrm{cl}(B(x,r))$ is contained in Ω, then (ii) gives that $s(x) \le I_{s,x,r}(x) = \mathcal{M}(s;x,r)$, which shows that s has the subharmonic mean value property on Ω, i.e. $s \in S(\Omega)$.

So far, we have shown (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii) $\Leftrightarrow$ (iv) $\Leftrightarrow$ (vi). Now we have just have to weave in (v). "(i) $\Rightarrow$ (v)" follows from (Theorem V.4)(ii). Now for "(v) $\Rightarrow$ (iii)": For $0 < r < r_0$, we define from Eq. (V.1) that

$$0 \le r^N(\mathcal{A}(s;x,r) - s(x)) = N\int_{[0,r)} \rho^N(\mathcal{M}(s;x,\rho) - \rho(x))\, d\lambda^1(\rho)$$

Thus, if we had that $\limsup_{r\downarrow 0} 1/r^2(\mathcal{M}(s;x,r) - s(x)) < 0$, then there were some $a > 0$ positive such that

$$\frac{1}{a^2}(\mathcal{M}(s;x,r) - s(x)) \leq -a < 0$$

for all sufficiently small values of r and hence $0 \leq -Na \int_{[0,r)} \rho^{N+2}\, d\lambda^1(\rho) < 0$, a contradiction. Therefore, (iii) must hold. □

The characterisation (vi) of subharmonic functions explains the name "subharmonic".

Corollary V.10: *Let $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ be given. If s_1, s_2 are subharmonic on Ω and $s_1 = s_2$ λ-almost everywhere on Ω, i.e. $\lambda^N\{x \in \Omega \mid s_1(x) \neq s_2(x)\} = 0$, then $s_1 = s_2$ on Ω.*

Proof: First, we claim that for a subharmonic function s on Ω and every $x \in \Omega$, it holds $\lim_{r\downarrow 0} \mathcal{A}(s;x,r) = s(x)$.

This can be seen as follow: By assertion (v) of Theorem V.8, we know that $\liminf_{r\downarrow 0} \mathcal{A}(s;x,r) \geq s(x)$ and by (Theorem V.4)(i), we know that

$$\limsup_{r\downarrow 0} \mathcal{A}(s;x,r) \leq \limsup_{y\to x} s(y) = s(x).$$

Thus, $\lim_{r\downarrow 0} \mathcal{A}(s;x,r)$ exists and equals $s(x)$.

Secondly, for two subharmonic functions s_1, s_2 on Ω with $s_1 = s_2$ λ^N-almost everywhere it follows for every $x \in \Omega$ and all $0 < r < r_0$ that

$$\mathcal{A}(s_1;x,r) = \mathcal{A}(s_2;x,r)$$

whenever $B(x,r_0) \subseteq \Omega$. By our first claim, this show that $s_1(x) = s_2(x)$. □

Corollary V.11: *Let $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ be an open set and let $s \in C^2(\Omega)$. Then s is subharmonic if and only if $\Delta s \geq 0$ on Ω.*

Proof: Since for all $x \in \Omega$ it holds

$$\lim_{r\downarrow 0} \frac{1}{r^2}(\mathcal{M}(s;x,r) - s(x)) = \frac{1}{2N}(\Delta s)(x),$$

the asserted equivalence is precisely the equivalence of assertions (i) and (iii) of Theorem V.8. □

In view of this result Corollary V.11, it is desirable to know how to approximate subharmonic functions by smooth functions. For this purpose, we have the following:

Theorem V.12: *Let $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ be open and let s be a subharmonic function on Ω. Furthermore, let $\varnothing \neq U \subseteq \mathbb{R}^N$ be open and bounded with $\mathrm{cl}(U) \subseteq \Omega$. Then, there is a sequence $(s_n)_{n\in\mathbb{N}}$ in $S(\Omega)\cap C^\infty(U)$ which is pointwise decreasing on U with $s(x) = \lim_{n\to\infty} s_n(x)$ for all $x \in U$.*

We record an important consequence:

Theorem V.13: *Let $\varnothing \neq \Omega_1, \Omega_2 \subseteq \mathbb{C}$ be domains and suppose that $f \in \mathcal{O}(\Omega_1)$ is non-constant with $f(\Omega_1) \subseteq \Omega_2$. Then, for a subharmonic function s on Ω_2 it holds that $s \circ f$ is subharmonic on Ω_1.*

Proof: Take any point $z_0 \in \Omega_1$ and choose a radius $r_2 > 0$ such that the open set $U_2 := D(f(z_0), r)$ satisfies $\mathrm{cl}(U_2) \subseteq \Omega_2$. Furthermore, take any radius $r_1 > 0$ such that $U_1 := D(z_0, r_1)$ satisfies $f^{-1}(U_2)$. It suffices to prove that $s \circ f|_{U_1}$ is subharmonic on U_1, since being subharmonic is a local property due to Theorem V.8 and since $z_0 \in \Omega_1$ was arbitrary.

By Theorem V.12 we find a sequence $(s_n)_{n\in\mathbb{N}}$ in of smooth subharmonic functions on U_2 which is pointwise decreasing on U_2 with limit $s|_{U_2}$. Therefore, $(s_n \circ f|_{U_1})_{n\in\mathbb{N}}$ is pointwise decreasing on U_1 with limit $s \circ f|_{U_1}$. Furthermore, $s_n \circ f$ is itself smooth on U_1. Because for all $z \in U_1$ it holds

$$\Delta(s_n \circ f)(z) = (\Delta s_n)(f(z))|f'(z)| \geq 0,$$

we know that $s_n \circ f$ is in fact subharmonic on U_1 (see Corollary V.11). By the Open Mapping Theorem (refer to Satz 13.5 of the lecture notes "Funtionentheorie I" from summer 2017), we know that $f(U_1)$ is open and thus, by (Theorem V.4)(iii), $s \not\equiv -\infty$ on $f(U_1)$ and so $s \circ f|_{U_1} \not\equiv -\infty$ on U_1. From exercise 2, assignment 3B, it follows that $s \circ f|_{U_1}$ is subharmonic as desired. □

The proof of Theorem V.12 uses convolution with smooth functions like in (Theorem III.5). We skip the details. We note however that the proof uses the following fact, which is a consequence of Fatou's Lemma and Fubini's Theorem:

Theorem V.14: *Let $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ be open and connected. Furthermore, let (Y, μ) be a σ-finite measure space. Suppose that $f\colon \Omega \times Y \to (-\infty, \infty]$ is measurable such that for all $y \in Y$ the function $f(\cdot, y)$ is superhamonic on Ω and*

there is μ-integrable functions $g\colon Y \to \mathbb{R}$ which satisfies for all $(x,y) \in \Omega \times Y$ that $f(x,y) \geq g(y)$. Then

$$u(x) := \int_Y f(x,y)\, d\mu(y)$$

declares a function $u\colon \Omega \to (-\infty, +\infty]$ which is either constantly $+\infty$ on Ω or superhamonic on Ω.

Remark V.15: Subharmonic functions have some remarkable applications in functional analysis.

Let $(A, \|\cdot\|)$ be a unital complex Banach algebra. For $a \in A$, the set $\sigma(a) := \{z \in \mathbb{C} \mid a - z\,\mathrm{id}_A \text{ not invertible in } A\}$ is called the *spectrum of A* and the real number $r(a) := \max\{|z| \mid z \in \sigma(a)\}$ is called the *spectral radius of a*. One can show that the following identity holds:

$$r(a) = \lim_{n\to\infty} \|a^n\|^{1/n} = \inf_{n\in\mathbb{N}} \|a^n\|^{1/n}.$$

Vesentini's Theorem says that, for every holomorphic function $f\colon \Omega \to A$ on a non-empty open set $\Omega \subseteq \mathbb{C}$, the function

$$s\colon \Omega \longrightarrow [-\infty, +\infty), \qquad z \longmapsto \log r(f(z))$$

is subharmonic, provided that $s \not\equiv -\infty$ on each connected component of Ω.

Using this remarkable fact, one can prove Johnson's Theorem which says that a surjective homomorphism $\Theta\colon A_1 \to A_2$ between complex Banach algebras A_1 and A_2 is automatically continuous if A_2 is semisimple (i.e. the intersection of all maximal right ideals in A_2 is $\{0\}$).

Chapter VI.

Riesz Measure

In Corollary V.11 we have seen that subharmonic C^2-functions have a non-negative Laplacian. Here, we generalise this observation to arbitrary subharmonic functions: The Laplacian of a subharmonic function, if understood in a distributional sense, yields then the Riesz Measure.

For a non-empty open subset $\varnothing \neq \Omega \subseteq \mathbb{R}^N$, we denote by $C_c(\Omega)$ the space of compactly supported continuous functions on Ω, i.e. of functions $f\colon \Omega \to \mathbb{R}$ such that $\operatorname{supp} f := \operatorname{cl}_\Omega(\{x \in \Omega \mid f(x) \neq 0\})$ is compact. By $C_c^\infty(\Omega)$ we denote the space of smooth compactly supported continuous functions on Ω.

Definition VI.1: Let $u\colon \Omega \to [-\infty, +\infty]$ be locally integrable on Ω. The *distributional Laplacian of* u is the linear functional

$$L_u\colon C_c^\infty(\Omega) \longrightarrow \mathbb{R}, \qquad L_U(f) := \int_\Omega u(x)(\Delta f)(x)\, d\lambda^N(x)\,.$$

Theorem VI.2: *Let $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ open.*

(i) *If $u \in C^2(\Omega)$, then for all $f \in C_c^\infty(\Omega)$ it holds*

$$L_u(f) = \int_\Omega (\Delta u)(x) f(x)\, d\lambda^N(x)\,.$$

(ii) *If h is harmonic on Ω, then $L_h \equiv 0$ on $C_c^\infty(\Omega)$.*

(iii) *If s is subharmonic on Ω, then L_s is a positive linear functional on $C_c^\infty(\Omega)$.*

Proof: (i) This follows from Greens Identity (see exercise 1, assignment 4B).

(ii) This is an immediate consequence of (i).

(iii) Take any $f \in C_c^\infty(\Omega)$ that is pointwise non-negative and let $\varnothing \neq U \subseteq \mathbb{R}^N$ be open such that $\operatorname{supp} f \subseteq U$. By Theorem V.12 there is a sequence $(s_n)_{n\in\mathbb{N}}$ of smooth subharmonic functions on U which is pointwise decreasing on U and convergent to s. By Corollary V.11, we know that for each natural number, Δs_n is pointwise non-negative on U and thus, by (i), we get that

$$\int_U s_n(x)(\Delta f)(x)\,d\lambda^N(x) = \int_U f(x)(\Delta s_n)(x)\,d\lambda^N(x) \geq 0.$$

By the monotone convergence of $(s_n(\Delta f)^\pm)_{n\in\mathbb{N}}$ on U, it follows that

$$\begin{aligned} L_s(f) &= \int_U s(x)(\Delta f)(x)\,d\lambda^N(x) \\ &= \lim_{n\to\infty} \int_U s_n(x)(\Delta f)(x)\,d\lambda^N(x) \geq 0 \end{aligned}$$

which we wanted to see. □

Reminder (Radon Measure): A measure μ on an open set $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ is called a *Radon measure*, if for all compact subsets $K \subseteq \Omega$ it holds $\mu(K) < \infty$.

Theorem VI.3: *Let $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ be an open subset and let s be a subharmonic function on Ω. Then there is a unique Radon measure μ_s on Ω, called the* Riesz measure associated to *s, such that for all $f \in C_c^\infty(\Omega)$ it holds*

$$L_s(f) = \int_\Omega f(x)\,d\mu_s(x)\,.$$

Proof: We show the assertion in three steps.

(1) *The distributional Laplacian L_s extends to a positive linear functional $\hat{L}_s\colon C_c(\Omega) \to \mathbb{R}$.*

By convolution with smooth functions, one can show that for all functions $f \in C_c(\Omega)$ there is a sequence of smooth compactly supported functions $(f_n)_{n\in\mathbb{N}}$ on Ω such that

$$\sup\{|f(x) - f_n(x)| \mid x \in \Omega\} \longrightarrow 0 \qquad (n \to \infty). \tag{VI.1}$$

Therefore, $(L_S(f_n))_{n\in\mathbb{N}}$ is a Cauchy sequence. One can show that the assignment $\hat{L}_s(f) := \lim_{n\to\infty} L_s(f_n)$ is independent of the approximating sequence and thus we obtain a well-defined linear extension of L_s to $C_c(\Omega)$.

(2) *By Riesz Representation Theorem, there is a (unique) Radon measure* μ_s *on* Ω *such that for all* $f \in C_c(\Omega)$ *it holds*

$$\hat{L}_s(f) = \int_\Omega f(x)\, d\mu_s(x)\,.$$

(3) *The obtained Radon measure is unique in the sense that we desire.*

Suppose μ_1, μ_2 were Radon measures on Ω such that for all $f \in C_c^\infty(\Omega)$ it holds

$$\int_\Omega f(x)\, d\mu_1(x) = \int_\Omega f(x)\, d\mu_2(x), \tag{VI.2}$$

then we had $\mu_1 = \mu_2$, because by Eq. (VI.1), we find that Eq. (VI.2) extends to holds for all $f \in C_c(\Omega)$. The uniqueness part of the Riesz Representation Theorem hence yields that $\mu_1 = \mu_2$. □

For a superharmonic function u on the open non-empty subset Ω of $\mathbb{R}^N$, we define the *Riesz measure* μ_u *associated to* u as the Riesz measure associated to $-u$, which is subharmonic on Ω.

Theorem VI.4: *Let* $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ *be open.*

(i) *The fundamental harmonic function* U_y *for* $\mathbb{R}^N$ *with pole at* $y \in \mathbb{R}^N$ *(see Definition II.4) extends by* $U_y(y) := \infty$ *to a superharmonic function. Its Riesz measure is given by* $\mu_{U_y} = a_N \delta_y$*, where* $a_N = \max\{1, N-2\} N \omega_N$ *and* δ_y *is the Dirac measure with atom* y*.*

(ii) *Let* μ *be a finite measure on* $\mathbb{R}^N$ *whose support*

$$\operatorname{supp}\mu := \{x \in \mathbb{R}^N \mid \forall\, \varepsilon > 0 : \mu(B(x,\varepsilon)) > 0\}$$

is compact. Then the potential ϕ_μ associated to μ*, which is defined by*

$$\phi_\mu(x) := \int_{\mathbb{R}^N} U_y(x)\, d\mu(y),$$

is itself superharmonic on $\mathbb{R}^N$ *and harmonic on* $\mathbb{R}^N - \operatorname{supp}\mu$*. The Riesz measure of* ϕ_μ *is given by* $\mu_{\phi_\mu} = a_N \mu$*.*

Proof: (i) By (Theorem II.3), U_y is harmonic on $\mathbb{R}^N - \{y\}$. Using the implication "(vi) ⇒ (i)" of Theorem V.8 it follows that U_y is superharmonic on $\mathbb{R}^n$. For $\mu_{U_y} = a_N \delta_y$, it suffices to show that for all compactly supportes smooth functions f on $\mathbb{R}^N$ it holds that $L_{U_y}(f) = a_N f(y)$. Let thus $f \in C_c^\infty(\mathbb{R})$

and let $r > 0$ be such that $\operatorname{supp} f \subseteq B(y,r)$. By Green's Identity (see Exercise 1 of assignment 4B), we get for $\varepsilon > 0$ that

$$\int_{A(y;\varepsilon,r)} U_y(x)(\Delta f)(x)\, d\lambda^N(x) = -\int_{\partial B(y,\varepsilon)} U_y(x)\langle \operatorname{grad} f(x), \nu(x)\rangle\, d\sigma(x) - \int_{\partial B(y,\varepsilon)} f(x)\langle \operatorname{grad} U_y(x), \nu(x)\rangle\, d\sigma(x)$$

with $\sigma := \sigma_{\partial B(y,\varepsilon)}$ and $\nu(x) = (x-y)/\|x-y\|$. Denote the first integral on the right-hand side by I_ε^1 and the second integral on the right-hand side by I_ε^2. Since for all $x \in \mathbb{R}^N - \{y\}$ it holds that

$$\operatorname{grad} U_y(x) = -\max\{1, N-2\}\frac{\nu(x)}{\|x-y\|^{N-1}},$$

we infer that, as $\varepsilon \downarrow 0$, $I_\varepsilon^2 = -a_N/\varepsilon^{N-1}\int_{\partial B(y,\varepsilon)} f(x)\, d\sigma(x) \to -a_N f(y)$. Since for all $x \in \partial B(y,\varepsilon)$ it holds that

$$U_y(x) = \begin{cases} -\log \varepsilon, & \text{if } N = 2, \\ \varepsilon^{2-N}, & \text{if } N \geq 3, \end{cases}$$

we infer (with the help of the Cauchy-Schwarz-inequality applied to the integrand)

$$|I_\varepsilon^1| \leq \max_{x \in \partial B(y,\varepsilon)} \|\operatorname{grad} f(x)\| \int_{\partial B(y,\varepsilon)} U_y(x)\, d\sigma(x) \longrightarrow 0 \quad \text{as } \varepsilon \downarrow 0,$$

and hence $I_\varepsilon^1 \to 0$ as $\varepsilon \downarrow 0$. In summary, we get that

$$L_{-U_y}(f) = -\lim_{\varepsilon \downarrow 0} \int_{A(y;\varepsilon,r)} U_y(x)(\Delta f)(x)\, d\lambda^N(x) = a_N f(y).$$

(ii) Due to Theorem V.8 it suffices to check for all $x_0 \in \mathbb{R}^N$ and $r > 0$ with $B(x_0,r) \cap \mathbb{R}^N - \operatorname{supp}\mu \neq \varnothing$ that $\Phi_\mu|_{B(x_0,r)}$ is superharmonic on $B(x_0,r)$. For every such ball $B(x_0,r)$, we apply (Theorem V.14) to the function

$$f\colon B(x_0,r) \times \Omega \longrightarrow (-\infty, +\infty], \qquad f(x,y) := U_y(x).$$

Because we know that $\Phi_\mu(x) < \infty$ for each $x \in \mathbb{R}^N - \operatorname{supp}\mu$, this gives that $\Phi_\mu|_{B(x_0,r)}$ is superharmonic on $B(x_0,r)$. If $B(x_0,r)$ is contained in $\mathbb{R}^N - \operatorname{supp}\mu$, then we can apply this argument to $-U_y$, which yields that $\Phi_\mu|_{B(x_0,r)}$ is in fact

harmonic on $B(x_0, r)$. In order to prove $\mu_{\Phi_\mu} = a_N \mu$, we must show that for all compactly supported smooth functions f on $\mathbb{R}^N$ it holds

$$L_{-\Phi_\mu}(f) = a_N \int_{\mathbb{R}^N} f(y)\, d\lambda^N(y)\,.$$

This follows with the help of Fubini's Theorem from (i):

$$\begin{aligned} L_{-\Phi_\mu}(f) &= -\int_{\mathbb{R}^N} \Phi_\mu(x)(\Delta f)(x)\, d\lambda^N(x) \\ &= -\int_{\mathbb{R}^N} \left(\int_{\mathbb{R}^N} U_y(x)\, d\mu(y) \right) (\Delta f)(x)\, d\lambda^N(x) \\ &= \int_{\mathbb{R}^N} \left(-\int_{\mathbb{R}^N} U_y(x)(\Delta f)(x)\, d\lambda^N(x) \right) d\mu(y) = \int_{\mathbb{R}^N} a_N f(y)\, d\mu(y), \end{aligned}$$

which we wanted to show. □

Remark VI.5: Let $\rho\colon \mathbb{R}^N \to \mathbb{R}$ be in $C_c^k(\mathbb{R}^N) = C_c(\mathbb{R}^N) \cap C^k(\mathbb{R}^N)$ for some $k \geq 1$. On $\mathbb{R}^N$ one can declare a function Φ via

$$\Phi(x) := \int_{\mathbb{R}^N} U_y(x)\rho(y)\, d\lambda^N(y),$$

which in fact belongs to $C^{k+1}(\mathbb{R}^N)$.

Furthermore, this function Φ solves *Poisson's equation* $\Delta\Phi = -a_N\rho$ on $\mathbb{R}^N$. Indeed, by Theorem VI.2(i), we have for all $f \in C_c^\infty(\mathbb{R}^N)$ that

$$L_\Phi(f) = \int_{\mathbb{R}^N} f(x)\Delta\Phi(x)\, d\lambda^N(x),$$

whereas, for the measures $\mu^\pm$ defined by $d\mu^\pm(y) := \rho^\pm(y)\, d\lambda^N(y)$, Theorem VI.4(ii) yields that

$$\begin{aligned} L_\Phi(f) &= L_{\Phi_{\mu^+}}(f) - L_{\Phi_{\mu^-}}(f) \\ &= -a_N \int_{\mathbb{R}^N} f(x)\, d\mu^+(x) + a_N \int_{\mathbb{R}^N} f(x)\, d\mu^-(x) \\ &= -a_N \int_{\mathbb{R}^N} f(x)\rho(x)\, d\lambda^N(x), \end{aligned}$$

where the first step relies on the decomposition $\Phi = \Phi_{\mu^+} - \Phi_{\mu^-}$ which is valid thanks to $\rho = \rho^+ - \rho^-$. Hence, in summary, for all $f \in C_c^\infty(\mathbb{R}^N)$ it holds

$$\int_{\mathbb{R}^N} f(x)\Big(\Delta\Phi(x) + a_N\rho(x)\Big)\, d\lambda^N(x) = 0,$$

which implies $\Delta\Phi(x) = -a_N\rho(x)$ for all $x \in \mathbb{R}^N$.

In particular, for $N = 3$, since $a_3 = 4\pi$, we get $\Delta\Phi = -4\pi\rho$. In the notation of Chapter 1, we have $\operatorname{grad}\Phi = -4\pi\varepsilon_0\vec{E}$, so that the latter identity yields $\operatorname{div}\vec{E} = \rho/\varepsilon_0$, which is known as the differential form of Gauss' Law from material on classical electrodynamics.

Chapter VII.

Logarithmic Potentials

From now on, we draw our attention to the case $N = 2$. We therefore identify $\mathbb{R}^2$ and $\mathbb{C}$. If μ is a finite (Borel) measure on $\mathbb{C}$ with compact support $K := \operatorname{supp}\mu$, then we refer to the potential

$$\Phi_\mu \colon \mathbb{C} \longrightarrow (-\infty, +\infty], \qquad z \longmapsto -\int_K \log|z-w|\, d\mu(w),$$

which was defined in Theorem VI.4(ii), as the *logarithmic potential associated to* μ. The logarithmic potential associated to μ is superharmonic on $\mathbb{C}$ and harmonic on $\mathbb{C} - K$.

Theorem VII.1: *Let μ be a finite measure on $\mathbb{C}$ with compact support K.*

(i) $\Phi_\mu(z) = -\mu(\mathbb{C})\log|z| + \log|z| + \mathcal{O}(1/|z|)$ *as* $z \to \infty$,

(ii) *Let $w_0 \in K$. Then*

$$\limsup_{z\to w_0} \Phi_\mu(z) = \limsup_{K\ni w\to w_0} \Phi_\mu(w).$$

Furthermore, if $\lim_{K\ni w\to w_0} \Phi_\mu(w) = \Phi_\mu(w_0)$, *then it holds*

$$\lim_{z\to w_0} \Phi_\mu(z) = \Phi_\mu(w_0).$$

(iii) *If there is a real bound M such that for all $w \in K$ it holds $\phi_\mu(w) \le M$, then it even holds $\phi_\mu(z) \le M$ for all $z \in \mathbb{C}$.*

Definition VII.2: Let μ be a finite measure on $\mathbb{C}$ with compact support K. We call

$$I(\mu) := \int_K \Phi_\mu(z)\, d\mu(z) = -\int_K \int_K \log|z-w|\, d\mu(z)\, d\mu(w)$$

the *energy of* μ.

Definition VII.3:

(i) A subset $E \subseteq \mathbb{C}$ is called *polar* if $I(\mu) = \infty$ for every finite measure $0 \neq \mu$ with compact support $\operatorname{supp} \mu \subseteq E$.

(ii) A property is said to hold *nearly everywhere on* $S \subseteq \mathbb{C}$ if it holds on $S - E$ for some Borel polar set E.

Theorem VII.4:

(i) *If μ is a finite measure on $\mathbb{C}$ with compact support satisfying $I(\mu) < \infty$, then $\mu(E) = 0$ for every Borel polar set.*

(ii) *Every Borel polar set $E \subseteq \mathbb{C}$ satisfies $\lambda^2(E) = 0$. In particular we find that "nearly everywhere" is stronger than "almost everywhere".*[1]

(iii) *A countable union of Borel polar sets is polar.*

Definition VII.5: Let $K \subseteq \mathbb{C}$ be compact. We denote by $P(K)$ the set of all probability measures on K (i.e. measure $\mu\colon \mathfrak{B}(K) \to [0,1]$ with $\mu(K) = 1$). If there exists $\nu \in P(K)$ such that $I(\nu) = \inf_{\mu \in P(K)} I(\mu)$, then ν is called an *equilibrium measure for* K.

Theorem VII.6: *Every compact set $K \subseteq \mathbb{C}$ has an equilibrium measure.*

Proof: We give a sketch of the proof. To achieve this, we list some facts:

(1) If $(\mu_n)_{n\in\mathbb{N}}$ is a sequence in $P(K)$ which is weak *-convergent to some $\mu \in B(K)$, i.e. for all $f \in C(K)$ it holds $\int_K f(z)\, d\mu_n(z) \to \int_K f(z)\, d\mu(z)$, then $\liminf_{n\to\infty} I(\mu_n) \geq I(\mu)$.

(2) Every sequence in $P(K)$ has a weak *-convergent subsequence.

Using these facts, we choose a sequence $(\mu_n)_{n\in\mathbb{N}}$ in $P(K)$ such that

$$I(\mu_n) \longrightarrow \inf_{\mu\in P(K)} I(\mu) \tag{VII.1}$$

as n tends to ∞. By (2), the sequence $(\mu_n)_{n\in\mathbb{N}}$ has a subsequence, say $(\mu_{n_k})_{k\in\mathbb{N}}$ which is weak *-convergent to some $\nu \in P(K)$. Then

$$\inf_{\mu\in P(K)} I(\mu) \leq I(\nu) \leq \liminf_{n\to\infty} I(\mu_{n_k}) = \inf_{\mu\in P(K)} I(\mu)$$

which shows that ν is an equilibrium measure for K. □

[1] One direction is clear from the above, the other direction can be shown but requires work.

Remark VII.7: If $K \subseteq \mathbb{C}$ is compact and not polar, then there is a unique equilibrium measure ν_K for K. Moreover, we have that $\operatorname{supp} \nu_K$ is contained in $\partial_e K$, where $\partial_e K$ denotes the *exterior boundary of* K, i.e. the boundary of the unbounded connected component of $\mathbb{C} - K$.

Theorem VII.8 (Frostman): *Let $K \subseteq \mathbb{C}$ be given and let ν be an equilibrium measure for K. Then, the following holds:*

(i) *For all $z \in \mathbb{C}$ we have $\Phi_\nu(z) \leq I(\nu)$,*

(ii) *There is some polar set $E \subseteq \partial K$ such that for all $z \in K - E$ it holds $\Phi_\nu(z) = I(\nu)$.*

Definition VII.9: Let $E \subseteq \mathbb{C}$ be any subset. We call

$$\operatorname{cap}(E) := \sup_{\mu \in P(E)} \exp(-I(\mu))$$

the *logarithmic capacity of* E. Here, $P(E)$ denotes the set of all Bore probability measures μ on $\mathbb{C}$ with compact support satisfying $\operatorname{supp} \mu \subseteq E$.

Let $K \subseteq \mathbb{C}$ be a compact set and let ν be an equilibrium measure for K. Note that then it holds $\operatorname{cap}(K) = \exp(-I(\nu))$.

Theorem VII.10: (i) *For $E \subseteq \mathbb{C}$, we have $\operatorname{cap}(E) = 0$ if and only if E is polar.*

(ii) *If $E_1 \subseteq E_2 \subseteq \mathbb{C}$, then $\operatorname{cap}(E_1) \leq \operatorname{cap}(E_2)$.*

(iii) *If $E \subseteq \mathbb{C}$ and α, β are complex numbers, then $\operatorname{cap}(\alpha E + \beta) = |\alpha| \operatorname{cap}(E)$.*

(iv) *If $K \subseteq \mathbb{C}$ is compact, then $\operatorname{cap}(K) = \operatorname{cap}(\partial_e K)$.*

(v) *For a compact set $K \subseteq \mathbb{C}$, we denote by $\Omega(K)$ the connected component of $(\mathbb{C} \cup \{\infty\}) - K$ which contains ∞. If $K_1, K_2 \subseteq \mathbb{C}$ are compact subsets and $f \colon \Omega(K_1) \to \Omega(K_2)$ is a meromorphic function satisfying $f(z) = z + O(1)$ as $z \to \infty$, then $\operatorname{cap}(K_2) \leq \operatorname{cap}(K_1)$. If f is even biholomorphic, then $\operatorname{cap}(K_2) = \operatorname{cap}(K_1)$.*

(vi) *If $K \subseteq \mathbb{C}$ is compact, then it holds*

$$\operatorname{cap}(K) \leq \frac{1}{2} \operatorname{diam}(K), \qquad \operatorname{cap}(K) \geq \sqrt{\frac{1}{\pi} \lambda^2(K)},$$

where $\operatorname{diam}(K) := \max\{|w_1 - w_2| \mid w_1, w_2 \in K\}$ and λ^2 denotes the Lebesgue measure on $\mathbb{C}$.

(vii) *If $K \subseteq \mathbb{C}$ is compact and $q(z) = \sum_{k=0}^{d} a_k z^k$ with $a_d \neq 0$ is a complex polynomial, then*

$$\operatorname{cap}(q^{-1}(K)) = \left(\frac{\operatorname{cap}(K)}{|a_d|}\right)^{1/d}.$$

Theorem VII.11 (Fekete-Szegö): *Let $K \subseteq \mathbb{C}$ be compact. Consider the sequence $(\delta_n(K))_{n\geq 2}$ of diameters of K, which was defined in exercise 2 of assignment 4B. Then $(\delta_n(K))_{n\geq 2}$ is convergent and the limit $\delta(K) := \lim_{n\to\infty} \delta_n(K)$ is given by $\delta(K) = \operatorname{cap}(K)$.*

Proof: From exercise 2(i) of assignment 4B, we know that $(\delta_n(K))_{n\geq 2}$ is decreasing and since $\delta_n(K) \geq 0$ for all $n \geq 2$, it follows that $(\delta_n(K))_{n\geq 2}$ is convergent. To make the proof easier to digest, we prove two smaller results first.

(1) *For all $n \geq 2$, it holds that $\delta_n(K) \geq \operatorname{cap}(K)$.* To see this, let $w_1, \ldots, w_n$ in K be given. By definition of $\delta_n(K)$ we have that

$$\frac{2}{n(n-1)} \sum_{1\leq i<j\leq n} \log|w_i - w_j| \leq \log \delta_n(K).$$

Hence, for every $\mu \in P(K)$, we get by integration of the latter inequality with respect to the product measure μ^n over K^n that

$$\frac{2}{n(n-1)} \sum_{1\leq i<j\leq n} \int_K \cdots \int_K \log|w_i - w_j| \, d\mu(w_1) \cdots d\mu(w_n) \leq \log \delta_n(K).$$

Since for each of the $n(n-1)/2$ possible choices of indices $1 \leq i < j \leq n$ it holds

$$\int_K \cdots \int_K \log|w_i - w_j| \, d\mu(w_1) \cdots d\mu(w_n) = \int_K \int_K \log|w_i - w_j| \, d\mu(w_i) \, d\mu(w_j) = -I(\mu),$$

we infer from the latter that $\exp(-I(\mu)) \leq \delta_n(K)$. Thus, it follows that $\operatorname{cap}(K) = \sup_{\mu\in P(K)} \exp(-I(\mu)) \leq \delta_n(K)$, as desired.

(2) *For each $n \geq 2$, let $w^{(n)} = (w_1^{(n)}, \ldots, w_n^{(n)})$ be a Fekete n-tuple for K and define $\mu_n \in P(K)$ by $\mu_n := \frac{1}{n}\sum_{i=1}^{n} \delta_{w_i^{(n)}}$. Let $(\mu_{n_k})_{k\in\mathbb{N}}$ be a subsequence of $(\mu_n)_{n\in\mathbb{N}}$ which is weak*-convergent to some $\nu \in P(K)$. Then $I(\nu) \leq -\log \delta(K)$.*

For $R > 0$, we set $\log_R(x) := \min\{\log(x), R\}$. Then, by monotone convergence, we find

$$I(\nu) = \lim_{R\to\infty} \int_K \int_K \log_R \frac{1}{|z-w|}\, d\nu(z)\, d\nu(w)$$

and thus, since $(\mu_{n_k})_{k\in\mathbb{N}}$ is weak*-convergent to ν, it follows

$$I(\nu) = \lim_{R\to\infty} \lim_{k\to\infty} \int_K \int_K \log_R \frac{1}{|z-w|}\, d\mu_{n_k}(z)\, d\mu_{n_k}(w)\,.$$

Next, we observe that

$$\begin{aligned}
&\int_K \int_K \log_R \frac{1}{|z-w|}\, d\mu_{n_k}(z)\, d\mu_{n_k}(w) \\
&\quad = \frac{1}{n_k^2} \sum_{i,j=1}^{n} \log_R \frac{1}{|w_i^{(n_k)} - w_j^{(n_k)}|} \\
&\quad = \frac{2}{n_k^2} \sum_{1\le i<j\le n}^{n} \log_R \frac{1}{|w_i^{(n_k)} - w_j^{(n_k)}|} + \frac{1}{n_k^2} n_k R \\
&\quad \le \frac{2}{n_k^2} \sum_{1\le i<j\le n}^{n} \log \frac{1}{|w_i^{(n_k)} - w_j^{(n_k)}|} + \frac{R}{n_k} = -\frac{n_k-1}{n_k} \log \delta_{n_k}(K) + \frac{R}{n_k}.
\end{aligned}$$

Hence, we deduce that

$$I(\nu) \le \lim_{R\to\infty} \lim_{k\to\infty} \Big(-\frac{n_k-1}{n_k} \log \delta_{n_k}(K) + \frac{R}{n_k} \Big) = -\log \delta(K),$$

as asserted.

Applying the results (1) and (2) in this order, we obtain that

$$\operatorname{cap}(K) \le \delta(K) \le \exp(-I(\nu)) \le \sup_{\mu\in P(K)} \exp(-I(\mu)) = \operatorname{cap}(K),$$

i.e. $\delta(K) = \operatorname{cap}(K)$, which proves the theorem.

Further, we see that ν must be an equilibrium measure for K. As there is a unique equilibrium measure ν_K for K in the case $\operatorname{cap}(K) > 0$, it follows from (1) that the sequence $(\mu_n)_{n\in\mathbb{N}}$ then has ν_K as its only limit point. Therefore, $(\mu_n)_{n\in\mathbb{N}}$ itself must be weak*-convergent to ν_K. □

Chapter VIII.

Uniform Approximation

Suppose that $K \subseteq \mathbb{C}$ is a compact set for which the complement $\mathbb{C} - K$ is connected. In this situation, Runge's Theorem says that every $f \in \mathcal{O}(\Omega)$, on some open set $\Omega \subseteq \mathbb{C}$ satisfying $K \subseteq \Omega$, can be approximated uniformly on K by (holomorphic) polynomials. We prove a quantitatie version of this result.

Theorem VIII.1 (Bernstein-Walsh): *Let $K \subseteq \mathbb{C}$ be compact, suppose that $\mathbb{C} - K$ is connected and let $\nu \in P(K)$ be an equilibrium measure for K. Suppose that $f \in \mathcal{O}(\Omega)$, where $K \subseteq \Omega \subseteq \mathbb{C}$ is open. Put*

$$\Theta := \begin{cases} \sup_{z \in (\mathbb{C} \cup \{\infty\}) - \Omega} \exp(\Phi_\nu(z) - I(\nu)), & \text{if } \operatorname{cap}(K) > 0, \\ 0, & \text{if } \operatorname{cap}(K) = 0. \end{cases}$$

Then $\Theta < 1$ and it holds $\limsup_{n \to \infty} d_n(f, K)^{1/n} \leq \Theta$, where

$$d_n(f, K) := \inf\{\|f - p\|_K \mid p \text{ holomorphic polynomial of degree at most } n\}.$$

The proof of this statement relies on the following result:

Theorem VIII.2 (Bernstein's Lemma): *In the situation of Theorem VIII.1 let $\operatorname{cap}(K) > 0$. Then it holds:*

(i) *If q is a polynomial with $n := \deg q \geq 1$, then for all $z \in \mathbb{C} - K$ it holds*

$$\left(\frac{|q(z)|}{\|q\|_K}\right)^n \leq \exp(-\Phi_\nu(z) + I(\nu)).$$

(ii) *If q is a Fekete polynomial for K with $n := \deg q \geq 2$, then for all $z \in \mathbb{C} - K$ it holds*

$$\left(\frac{|q(z)|}{\|q\|_K}\right)^{1/n} \geq \exp(-\Phi_\nu(z) + I(\nu)) \left(\frac{\operatorname{cap}(K)}{\delta_n(K)}\right)^{\tau(z,\infty)},$$

where $\tau := \tau_{\Omega(K)}$ is the Harnack distance on $\Omega(K) = (\mathbb{C} \cup \{\infty\}) - K$.

Remark VIII.3: (i) Let $\varnothing \neq \Omega \subseteq \mathbb{R}^N$ be open and connected. Building on exercise 1 of assignment 3A, we define for $x, y \in \Omega$ that

$$\tau_\Omega(x,y) := \inf\{\tau > 0 \mid \forall\, u \in H_+(\Omega) : \tau^{-1}u(x) \leq u(y) \leq \tau u(x)\}$$

We call $\tau_\Omega \colon \Omega \to \Omega \to [1,\infty)$ the Haranck distance on Ω. One can show that $\log \tau_\Omega \colon \Omega \times \Omega \to [0,\infty)$ is a continuous semimetric on Ω.

(ii) Due to (Theorem V.13), one can extend the notion of sub-, super and harmonic functions to Riemann surfaces and in particular to the Riemann sphere $\mathbb{C} \cup \{\infty\}$. For instance, the Harnack distance and the maximum principle (Theorems 3.8 and 5.7) remain true.

Proof: (i) Without loss of generality, we may assume that q is monic (i.e. $q(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0$). On $\mathbb{C} - K$ we declare a subharmonic function u via

$$u(z) := \frac{1}{n}\log|q(z)| - \frac{1}{n}\log\|q\|_K + \Phi_\nu(z) - I(\nu).$$

Note that $z \mapsto 1/n \log|q(z)|$ is subharmonic on $\mathbb{C} - K$, refer to exercise 1 of assignment 1B and (Theorem V.8). Since $\Phi_\nu(z) = -\log|z| + \mathcal{O}(1/|z|)$ (see Theorem VII.1) and $1/n \log|q(z)| = \log|z| + o(1)$ as $z \to \infty$, we see that u extends to a subharmonic function on $\Omega(K)$ by $u(\infty) := -1/n \log\|q\|_K$. Now, for every $w \in \partial K$, we get by (Theorem VII.8)(i) that

$$\limsup_{z \to w} u(z) \leq \frac{1}{n}\log|q(w)| - \frac{1}{n}\log\|q\|_K.$$

By (Theorem V.7)(iii) (see also (Remark VIII.3)(iii)), if follows that $u \leq 0$ on $\Omega(K)$, which implies the assertion.

(ii) Note that all zeros of q lie in K, thus, u is harmonic on $\Omega(K)$ and $u \leq 0$. Hence $-u \in H_+(\Omega(K))$ and (Remark VIII.3)(i) gives that for all $z \in \Omega(K)$ it holds

$$u(z) \geq \tau_{\Omega(K)}(z,\infty)u(\infty).$$

By exercise 2(ii) of assignment 4B, we have the estimate

$$u(\infty) = -I(\nu) - \frac{1}{n}\log\|q\|_K \geq -I(\nu) - \log\delta_n(K) = \log\left(\frac{\operatorname{cap}(K)}{\delta_n(K)}\right).$$

Putting this together, we obtain the result. □

Proof (of Theorem VIII.2): Suppose that $\mathrm{cap}(K) > 0$. Let Γ be a closed contour in $\Omega - K$ such that $\mathrm{Ind}_\gamma(w) = 1$ for all $w \in K$ and $\mathrm{Ind}_\Gamma(z) = 0$ for all $z \in \mathbb{C} - \Omega$. By the global version of Cauchy's Integral Formula (see Satz 7.12 in the Funktionentheorie I lecture notes), we have for all $w \in K$ that

$$f(w) = \frac{1}{2\pi \mathrm{i}} \int_\Gamma \frac{f(z)}{z - w}\, dz\,. \tag{VIII.1}$$

For $n \geq 2$, let q_n be a Fekete polynomial of degree n for K and put

$$p_n(w) := \frac{1}{2\pi \mathrm{i}} \int_\Gamma \frac{f(z)}{q_n(z)} \frac{q_n(w) - q_n(z)}{w - z}\, dz \tag{VIII.2}$$

Then p_n is a polynomial with $\deg p_n \leq n - 1$. From Eq. (VIII.1), we deduce that for all $w \in K$ it holds

$$f(w) - p_n(w) = \frac{1}{2\pi \mathrm{i}} \int_\Gamma \frac{f(z)}{z - w} \frac{q_n(w)}{q_n(z)}\, dz,$$

hence $d_n(f, K) \leq \|f - p_n\|_K \leq C\|q_n\|_K / \inf_{z\in\Gamma}|q_n(z)|$, where C is the constant $C := \|f\|_\Gamma L(\Gamma)/[2\pi \cdot \mathrm{dist}(\Gamma, K)]$. By (Theorem VIII.2)(iii) we get for all $z \in \Gamma$ that

$$\left(\frac{\|q_n\|_K}{|q_n(z)|}\right)^{1/n} \leq \exp(\Phi_\nu(z) - I(\nu)) \left(\frac{\delta_n(K)}{\mathrm{cap}(K)}\right)^{\tau_{\Omega(K)}(z,\infty)} \leq \Theta_\Gamma \left(\frac{\delta_n(K)}{\mathrm{cap}(K)}\right)^{\alpha},$$

where $\Theta_\Gamma := \sup\{\exp(\Phi_\nu(z) - I(\nu)) \mid z \in \Gamma\}$ and $\alpha := \sup\{\tau_{\Omega(K)}(z, \infty) \mid z \in \Gamma\}$. Hence, due to (Theorem VII.11), it holds

$$\limsup_{n\to\infty} d_n(f, K)^{1/n} \leq \limsup_{n\to\infty} C^{1/n} \Theta_\Gamma \left(\frac{\delta_n(K)}{\mathrm{cap}(K)}\right)^{\alpha} = \Theta_\Gamma.$$

Finally, we note that for all $\varepsilon > 0$, there is a contour Γ as above such that $0 \leq \Theta_\Gamma - \Theta < \varepsilon$. This proves the fact $\limsup_{n\to\infty} d_n(f, K)^{1/n} \leq \Theta$, if K has positive capacity.

Note that $\Theta < 1$, because otherwise, there was a $z_0 \in \mathbb{C} - K$ such that $\Phi_\nu(z_0) = I(\nu)$ and thus, by Theorem VII.7(i), z_0 was a local maximum so that $\Phi_\nu \equiv I(\nu)$ on $\mathbb{C} - K$ by Theorem V.7(ii). But this contradicted Theorem VII.1(i), as $I(\nu) < \infty$.

The assertion in the case $\mathrm{cap}(K) = 0$ follows by approximation of K with a decreasing sequence $(K_k)_{k\in\mathbb{N}}$ of non-polar compact subsets of Ω satisfying $K = \bigcap_{k=1}^\infty K_k$ from the already proved case. □

Remark VIII.4: We notice that the polynomials p_n defined in Eq. (VIII.2) satisfy for $1 \leq j \leq n$ that

$$p_n(w_j) = \frac{1}{2\pi \mathrm{i}} \int_\Gamma \frac{f(z)}{q_n(z)} \frac{q_n(w_j) - q_n(z)}{w_j - z}\, dz = \frac{1}{2\pi \mathrm{i}} \int_\Gamma \frac{f(z)}{z - w_j}\, dz = f(w_j),$$

where, in the second step, we have used that $q_n(w_j) = 0$, and, in the last step, Cauchy's integral formula as formulated in Eq. (VIII.1). In other words, p_n solves the following interpolation problem:

> Find a holomorphic complex polynomial p with $\deg p \leq n-1$ such that for $1 \leq j \leq n$ it holds $p(w_j) = f(w_j)$.

Note that if $w_1, \ldots, w_n$ are all distinct, then p_n is the unique solution of this interpolation problem. In this case, one can use the so-called *Lagrange polynomials* to find an explicit expression for p_n.

Among all holomorphic complex polynomials p satisfying $\deg p \leq n$, there is always at least one *best approximation* p_* to f, i.e. p_* satisfies the condition $d_n(f, K) = \|f - p_*\|_K$. In general, the polynomials p_n defined in Eq. (VIII.2) do not provide best approximations to f. Therefore, it seems plausible that a better choice of p_n might lead to better results about the asymptotic behaviour of $d_n(f, K)$ as $n \to \infty$. However, one can show that for $n \geq 2$ it always holds

$$\|f - p_n\|_K \leq (n+1) d_n(f, K).$$

Thus, we see that $\limsup_{n\to\infty} d_n(f, K)^{1/n} \leq \Theta$, namely the conclusion of (Theorem VIII.1), holds if and only if $\limsup_{n\to\infty} \|f - p_n\|_K^{1/n} \leq \Theta$.

Example VIII.5: Fix $z_0 \in \mathbb{C}$ and $r_0 > 0$ and put $K := \mathrm{cl}(D(z_0, r_0))$. The capacity of K can be shown to be $\mathrm{cap}(K) = r_0$ and the (unique) equilibrium measure ν_K is given by $\nu_K = (2\pi r_0)^{-1} \sigma_{\partial D(z_0, r_0)}$. One finds that the associated logarithmic potential Φ_{ν_K} is of the form

$$\Phi_{\nu_K}(z) = \begin{cases} \log 1/r_0, & \text{if } |z - z_0| \leq r_0, \\ \log 1/|z - z_0|, & \text{if } |z - z_0| > r_0. \end{cases}$$

Now, for any $r > r > 0$, we consider $\Omega := D(z_0, r)$. Then,

$$\Theta = \sup\{\exp(\Phi_{\nu_K}(z) - I(\nu)) \mid z \in (\mathbb{C} \cup \{\infty\}) - \Omega\} = \frac{r_0}{r}.$$

Hence, Theorem VIII.1 asserts that $\limsup_{n\to\infty} d_n(f,K)^{1/n} \leq \Theta$ for every $f \in \mathcal{O}(\Omega)$. This is in keeping with the rate of approximation of f by its Taylor polynomials at the point z_0. In fact, if we put

$$T_n(w) := \sum_{k=0}^{n} \frac{f^{(k)}(z_0)}{k!}(w-z_0)^k$$

for every integer $n \geq 0$, then Cauchy's Integral Formula yields for every $r_0 < \rho < r$ and all $w \in K$ that

$$\begin{aligned} f(w) - T_n(w) &= \frac{1}{2\pi\mathrm{i}} \int_{\partial D(z_0,\rho)} f(\zeta) \left(\frac{1}{\zeta - w} - \sum_{k=0}^{n} \frac{(w-z_0)^k}{(\zeta - z_0)^{k+1}} \right) d\zeta \\ &= \frac{1}{2\pi\mathrm{i}} \int_{\partial D(z_0,\rho)} f(\zeta) \sum_{k=n+1}^{\infty} \frac{(w-z_0)^k}{(\zeta - z_0)^{k+1}}\, d\zeta\,. \end{aligned}$$

We infer from the latter that $\|f - T_n\|_K \leq \|f\|_{\partial D(z_0,\rho)} (1 - \frac{r_0}{\rho})^{-1} (\frac{r_0}{\rho})^{n+1}$, which yields

$$\limsup_{n\to\infty} \|f - T_n\|_K^{1/n} \leq \frac{r_0}{\rho}.$$

As $r_0 < \rho < r$ was arbitrary, we can let $\rho \uparrow r$ and obtain the estimate $\limsup_{n\to\infty} \|f - T_n\|_K^{1/n} \leq \Theta$.

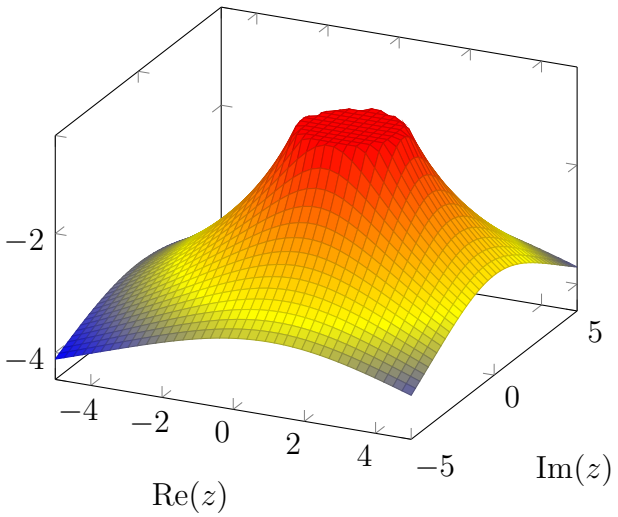


Figure VIII.1.: Graph of the potential Φ_{ν_K} for the equilibrium measure ν_K for $K = \mathrm{cl}(D(z_0, r_0))$ with $z_0 = 1$ and $r_0 = 2$, see Example VIII.5.

Example VIII.6: For the interval $K = [-1, 1]$, one can show that $\operatorname{cap}(K) = 1/2$ and that the (unique) equilibrium measure is given by $d\nu_K(x) = (1 - x^2)^{-2}\, dx$. Furthermore, one obtains that

$$\Phi_{\nu_K}(z) = \begin{cases} \log(2), & \text{if } z \in [-1, 1], \\ \log(2) - \log|z + (z^2 - 1)^{1/2}|, & \text{if } z \in \mathbb{C} - [-1, 1]. \end{cases}$$

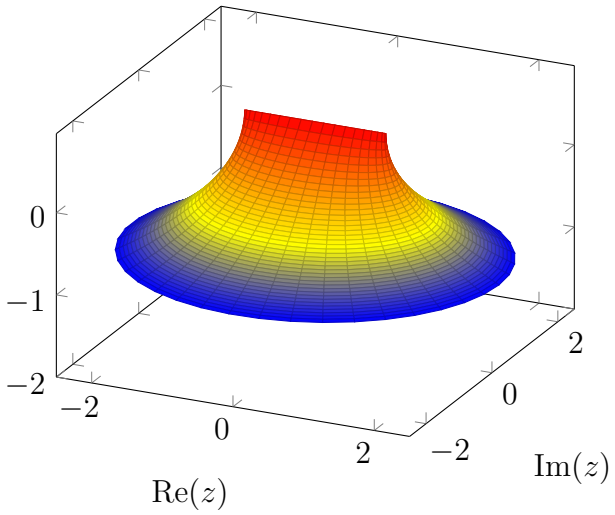


Figure VIII.2.: Graph of the potential Φ_{ν_K} for the equilibrium measure ν_K for $K = [-1, 1]$, see Example VIII.6.